

# A Filtered-Label Calibrated XGBoost Framework with Walk-Forward Validation for Robust Bitcoin Direction Prediction

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## Abstract

Daily directional prediction of Bitcoin (BTC) is widely regarded as approximately a random walk, with most published classifiers attaining accuracy close to 50%. We address this problem by filtering the training labels to retain only days whose forward magnitude exceeds a static or volatility-adaptive threshold, training a calibrated XGBoost classifier on a 120+-feature daily matrix under calendar-refit walk-forward validation, and evaluating performance over a 180-cell experimental design (timing  $\times$  cadence  $\times$  feature version  $\times$  NaN policy  $\times$  holdout). The leading 730-day returns range from +120% to +255% across the leaderboard; we therefore subject the leaderboard to four complementary robustness procedures: Bonferroni and Holm FWER corrections, the Deflated Sharpe Ratio of Bailey and López de Prado (2014), the Probability of Backtest Overfitting via Combinatorially Symmetric Cross-Validation (CSCV), and slippage sensitivity. Three of 126 (cell  $\times$  window) selections pass strict Bonferroni control at  $\alpha = 0.05$  — all under UTC 21:30 timing with semi-monthly retraining cadence — and only one cell (utc2130\_sm\_v5) attains PBO  $< 0.20$ . The UTC 00:00 sub-population, including its leading +231% cell, fails every strict correction and degrades by approximately 41 percentage points under a realistic 20 bps slippage assumption. From the surviving cells we construct four SPLIT-CAPITAL ensembles (Top1 / Top2 / Top3 / Top5); Top2 is Calmar-optimal at 9.83, halving the single-cell drawdown to  $-7.44\%$  while preserving +199.85% return. The four ensembles are deployed in parallel within a production service (coin\_service\_utc2130) so that end users can examine the full risk profile prior to selection.

**Keywords:** *bitcoin, direction prediction, gradient boosting, probability calibration, isotonic regression, walk-forward validation, calendar-refit cadence, dynamic labels, UTC alignment, cryptocurrency, quantitative trading*

## Introduction

Bitcoin’s price dynamics have been the subject of an extensive empirical literature. Despite the abundance of high-frequency price data, perpetual-futures positioning, on-chain analytics, and high-quality cross-asset macroeconomic series, supervised classifiers designed to predict the next-day direction of BTC rarely attain accuracies above 55% [11, 12]. A recurring finding is that BTC daily returns are approximately a martingale, and that the residual autocorrelation is regime-dependent and unstable.

We argue that the daily-direction task, as conventionally formulated, conflates two qualitatively distinct event classes: small intraday fluctuations whose sign is dominated by microstructure noise, and larger directional moves driven by macroeconomic shocks, derivatives positioning, sentiment regimes, and cross-asset risk-on/risk-off rotation. Pooling the two into a single binary label compels the classifier to fit both signal and noise on the same loss surface.

We therefore propose to filter the training set so as to retain only the larger moves. Specifically, given a threshold  $\tau$ , we retain training rows whose next-day absolute return exceeds  $\tau$  and discard the remainder:

$$y_t = 1 \text{ if } r_{t+1} > +\tau, \quad y_t = 0 \text{ if } r_{t+1} < -\tau, \quad y_t = NaN \text{ otherwise}$$

*Equation (1). Filtered binary target. NaN-labelled days are excluded from training but retained in the inference set.*

In addition to this label-filtering scheme, we introduce a second target mode in which the per-day filter threshold scales with realised volatility,  $\tau_t = k \cdot \hat{\sigma}_{30}(r)$ , so that the model is trained on a stable proportion of moves across regimes. This label design is then combined with (a) multi-timeframe feature engineering, (b) dedicated regime indicators, (c) cross-asset, futures, and sentiment features, (d) post-hoc isotonic probability calibration applied in a walk-forward manner, and (e) a structured search across UTC-bar timing, calendar-refit cadence, feature version, and

NaN-handling policy. The resulting framework yields high-conviction BUY/HOLD/SELL signals across a broad region of the threshold grid.

Our contributions are as follows. (i) We introduce a 180-cell experimental design over (timing  $\times$  cadence  $\times$  feature-version  $\times$  NaN-policy  $\times$  horizon), in which each cell is itself a  $22 \times 21 = 462$  (label  $\times$  probability) sweep — combining 11 static values and 11 dynamic  $k$  values across 21 BUY-probability thresholds — yielding 83,160 backtests in total. (ii) We subject the resulting leaderboard to four complementary statistical-robustness procedures — Bonferroni and Holm FWER, the Deflated Sharpe Ratio of Bailey and López de Prado (2014), Combinatorially Symmetric Cross-Validation (CSCV) for the Probability of Backtest Overfitting, and slippage sensitivity. Of 126 (cell  $\times$  window) selections, only three pass strict Bonferroni control at  $\alpha = 0.05$ , all under UTC 21:30 timing with semi-monthly retraining cadence, and a single cell (utc2130\_sm\_v5) attains PBO  $< 0.20$ . (iii) We document a counterintuitive timing effect: UTC 00:00 dominates UTC 21:30 in raw pairwise return comparison despite carrying a one-day macroeconomic lag — indicating that zero-lag access to same-day macroeconomic data leaks price-driven information into the feature vector — yet this raw advantage is reversed by the robustness analyses in (ii) and by realistic slippage assumptions, motivating UTC 21:30 SM-cadence as the production timing. (iv) We present a complete ablation over calendar-refit cadences (monthly, semi-monthly, weekly Sunday) and feature versions (v0–v6), demonstrating that computationally inexpensive monthly refits with the v0 baseline remain competitive on the raw leaderboard, although only UTC 21:30 SM-cadence variants survive the joint robustness filter. (v) From the surviving cells we construct four SPLIT-CAPITAL ensembles — Top1, Top2, Top3, and Top5 — which span the trade-off between absolute return, drawdown control, signal frequency, and component diversification. The Calmar-optimal Top2 ensemble preserves +199.85% over 730 days while halving the single-cell drawdown to  $-7.44\%$ ; all four ensembles are deployed in parallel within a production service so that users can examine the full risk profile prior to selection.

## Related Work

A substantial body of work has applied gradient-boosted trees [2] and recurrent neural networks to direction prediction in cryptocurrency markets [11, 12]. Reported accuracies for daily BTC direction typically lie in the 48–55% range, and the strongest results often arise either from sub-daily horizons, where microstructure features are informative, or from heterogeneous feature integration. Jaquart, Dann, and Weinhardt [22] benchmark recurrent, gradient-boosting, and feed-forward classifiers on prediction horizons from one to sixty minutes and report that technical features remain dominant for most methods, followed by selected blockchain- and sentiment-based features — a result that motivates the multi-block feature design of §4. Mallqui and Fernandes [26] explicitly target the direction and OHLC of daily BTC and find that ensembles of recurrent and tree-based models materially outperform single-model baselines, paralleling the SPLIT-CAPITAL ensembles of §8. Zhang et al. [27] demonstrate that attention-augmented convolutional architectures improve cryptocurrency price prediction, while Chang et al. [28] and Koo and Kim [29] decompose the price series into frequency or cluster components prior to forecasting — signal-decomposition approaches that address the same heterogeneous-event problem we address by filtered-label thresholding (§4.1).

On the trading-strategy side, Hajek, Hikkerova, and Sahut [23] show that an ensemble combining sentiment-derived features with machine-learning classifiers materially improves daily BTC prediction, directly supporting Block F (Fear & Greed) of our feature catalogue and the SPLIT-CAPITAL ensembles of §8. Liu et al. [24] forecast more than three thousand cryptocurrency returns with machine learning and report a Sharpe ratio of 1.95 on an equal-weighted long portfolio, providing a useful comparator for the risk-adjusted performance reported in §7. Sebastião and Godinho [25] explicitly evaluate machine-learning trading strategies under changing market conditions, paralleling the four-window consistency check of §7.20 and the regime decomposition of §7.14. Filtered-label and triple-barrier methods, popularised by López de Prado [1], address the noise problem by redefining the target in terms of realised economic events (price barriers, time horizons) rather than calendar bars; our static and dynamic filtered targets can be regarded as special cases of this formulation, with the time horizon fixed at one day and the barrier either constant (static) or volatility-adaptive (dynamic). Probability calibration of tree ensembles via isotonic regression [5, 6] and Platt scaling [4] has been extensively studied in the tabular machine-learning literature; its specific impact on threshold-based trading rules, however, remains comparatively under-investigated.

## Data

We assemble a heterogeneous data set drawn entirely from publicly available sources, none of which require commercial licences. BTC OHLCV data is obtained from Bitstamp (BTC/USD, from 2011-08-18) and Binance (BTC/USDT, from 2017-08-17) and concatenated into a continuous series. Macroeconomic variables are retrieved from Yahoo Finance: the

U.S. Dollar Index (DX-Y.NYB), front-month gold futures (GC=F), the S&P 500 Index ( $\hat{G}SPC$ ), the CBOE Volatility Index ( $\hat{VIX}$ ), the 10-year Treasury yield ( $\hat{TNX}$ ), and Ethereum (ETH-USD). Binance USDT-margined perpetual funding rates are aggregated to daily mean, maximum, and minimum from 2019-09-08 onwards. The perpetual-versus-spot basis is computed from /fapi mark-price klines and /api spot klines. The Fear & Greed Index is obtained from alternative.me (daily, from 2018-02-01). Coin Metrics community-edition daily metrics (active addresses, transaction counts, hash rate) supply the v3+ feature versions, and the FRED H.6 release provides the M2 money stock for the v4+ versions.

For on-chain analytics that require a paid API (true MVRV, NUPL, SOPR, exchange netflows) we substitute a price-based proxy in which the Realised Price is approximated by a 365-day moving average of closing prices.

All series can be aligned to one of two daily UTC bar closes. The *UTC 00:00* alignment corresponds to the conventional midnight close: the day- $t$  feature vector contains all external feeds with a one-day lag (e.g., the day- $(t-1)$  U.S. equity close, the day- $(t-1)$  Fear-and-Greed fix). The *UTC 21:30* alignment shifts the daily bar to 21:30 UTC (= 06:30 KST on the following day), which is subsequent to every same-day publication on the source feeds; under this alignment the feature vector contains zero-lag macroeconomic and sentiment readings. We evaluate both alignments and find that UTC 00:00 in fact dominates UTC 21:30 (§7.6) — a counterintuitive but interpretable result, since the same-day macroeconomic reading at 21:30 also encodes the U.S. session’s price action, which subsequently leaks into the day- $t$  target.

External features that are unavailable for older portions of the daily index (e.g., Binance funding prior to 2019-09) are left as NaN and handled natively by XGBoost’s missing-value-aware splits. To limit the prevalence of NaN-heavy periods in the training set, all variants restrict the training rows to those on or after 2019-12-01, by which date the 30- and 60-day rolling z-scores of the external features have stabilised.

## Feature Engineering

All variants share a common 99-feature baseline (the v0 set). The complete catalogue, including descriptions, formulas, and data sources, is reproduced in Appendix A. The 99 columns are organised into eleven blocks summarised below; for brevity we describe only the additions that differentiate v3, v4, v5, and v6 from v0.

Block A (Daily / Weekly / Monthly TA, 27 features). Per timeframe: close, one-bar return, RSI14 [7], MACD histogram [8], Bollinger band position [9], fast/slow MA ratio, return volatility, range %, volume z-score. Higher-timeframe values are joined to the daily index with as-of (backward) merges:

$$x_t^{(higher)} = x_{s^*}^{(higher)}, \quad s^* = \max\{s : t_s \leq t\}$$

*Equation (2). As-of (backward) join used to align the weekly/monthly timeframes to the daily index.*

Block B (Window summaries, 9 features). Per timeframe, the slope, cumulative return, and maximum drawdown of the trailing  $N$  bars (180 daily, 26 weekly, 6 monthly).

Block C (15-minute intraday, 6 features). Vectorised 24h and 5d roll-ups of the 15-minute series projected onto the daily index.

Block D (Sideways / regime, 11 features). Kaufman’s Efficiency Ratio [10], two volatility-adaptive sideways ratios, two Choppiness Index horizons, the 20-day Bollinger band width, the 30-day high-low range, and a streak counter for consecutive sideways days.

Block E (Macro, 28 features). For each Yahoo series we compute 1-day and 5-day returns, 20-day return volatility, and a 60-day z-score; we add the ETH/BTC ratio and its 5-day return.

Block F (Futures + sentiment, 12 features). Funding-rate mean / max / min, 7-day funding moving average, 30-day funding z-score, basis, 5-day basis MA, 30-day basis z-score, and five Fear & Greed features [13].

Block G (Calendar + MVRV proxy, 6 features). Day of week, month, days to and since the nearest BTC halving event, and the price-based MVRV-Z proxy [14].

## Feature Versions (v0 $\rightarrow$ v6)

We define five progressively enriched feature sets, each evaluated as a distinct variant of the production sweep:

| version | definition                                                                                                                                                                                                                                                                      | extra source    |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| v0      | Baseline 99 features (Blocks A–G).                                                                                                                                                                                                                                              | —               |
| v3      | v0 + Coin Metrics on-chain (active-address momentum, tx-per-address, hash-rate ribbon, hash-rate 7-day change), cross-asset 30-day rolling correlations (BTC vs SPX / DXY / Gold / ETH), funding skew & velocity, vol-of-vol, vol-regime z-score; redundant v0 columns dropped. | Coin Metrics CE |
| v4      | v3 + FRED M2 money stock (level, 13/26/52-week pct change, 60-day z-score) + restored 60-day z-scores of the four macro series that v3 dropped.                                                                                                                                 | FRED H.6        |
| v5      | v4 pruned to the top-60 features by mean walk-forward importance.                                                                                                                                                                                                               | —               |
| v6      | v4 features with tighter XGBoost regularisation (reg_alpha = 0.8, reg_lambda = 3.0, max_depth = 3, min_child_weight = 10, colsample_bytree = 0.5).                                                                                                                              | —               |

Table 1. Feature version definitions.

## Methodology

### Filtered Binary Targets — Static and Dynamic

**Static target** (Equation 1). We sweep  $\{0.010, 0.011, \dots, 0.020\}$  (eleven values). **Dynamic target**. The threshold is defined as a per-day function of realised volatility:

$$\tau_t = k \cdot \hat{\sigma}_{30}(r_t), \quad y_t = 1 \text{ if } r_{t+1} > +\tau_t, \quad y_t = 0 \text{ if } r_{t+1} < -\tau_t, \quad y_t = \text{NaN} \text{ otherwise}$$

Equation (3). Volatility-adaptive (dynamic) filtered binary target, with  $\hat{\sigma}_{30}(r)$  the 30-day rolling standard deviation of one-bar returns.

We sweep  $k \in \{1.0, 1.1, \dots, 2.0\}$  (eleven values). NaN-labelled rows are excluded from training but retained in the inference set, so that calibrated probabilities can be produced for every day.

### Classifier and Calibration

The base learner is a 400-tree XGBoost classifier [2] with maximum depth 4, learning rate 0.04, min\_child\_weight 6, subsample 0.9, colsample\_bytree 0.8,  $\gamma = 0.2$ , and L1/L2 regularisation parameters  $\alpha = 0.2$  and  $\beta = 1.5$ . The classifier is wrapped in scikit-learn’s [3] CalibratedClassifierCV using isotonic regression [5] with 3-fold internal cross-validation. The v6 variant overrides  $\gamma$ ,  $\alpha$ , max\_depth, min\_child\_weight, and colsample\_bytree as specified in §4.1.

### Walk-Forward with Calendar-Refit Cadence

Predictions are produced by walk-forward refits on a calendar schedule rather than at fixed N-day intervals. Three cadences are evaluated:

| cadence | refit days                          | approximate refit count  |
|---------|-------------------------------------|--------------------------|
| M       | First day of each calendar month    | 24 fits per 730d window  |
| SM      | 1st and 16th of each calendar month | 48 fits per 730d window  |
| WSUN    | Every Sunday                        | 104 fits per 730d window |

Table 2. Calendar-refit cadences.

Between successive refits, the cached pickled model scores incoming feature vectors. The refit boundary is aligned with calendar dates so that operational retraining can be scheduled deterministically (for example, via a cron job on the first of each month). The cold-start period comprises all data preceding the holdout cutoff.

### Three-State Decision Rule

Given the calibrated probability  $p_t$ , we assign one of three discrete signals:

$$\text{signal}_t = \text{BUY} \text{ if } p_t \geq \tau_{\text{buy}}, \quad \text{SELL} \text{ if } p_t < \tau_{\text{sell}}, \quad \text{HOLD} \text{ otherwise}$$

Equation (4). Three-state decision rule.

We sweep  $\_buy \in \{0.50, 0.51, \dots, 0.70\}$  (twenty-one values) and fix  $\_sell = 0.50$ . Strategy returns are compounded, with a per-side transaction cost of 10 bps.

## Experiments

The full design crosses five axes:

| axis            | levels                 | count    |
|-----------------|------------------------|----------|
| Timing          | UTC 00:00 vs UTC 21:30 | 2 levels |
| Cadence         | M / SM / WSUN          | 3 levels |
| Feature version | v0 / v3 / v4 / v5 / v6 | 5 levels |
| NaN policy      | raw / close / complete | 3 levels |
| Holdout horizon | 365-day / 730-day      | 2 levels |

Table 3. The  $2 \times 3 \times 5 \times 3 \times 2 = 180$ -cell experimental design. Each cell is itself a  $22 \times 21$  (label  $\times$  prob) sweep (11 static + 11 dynamic  $k$ , 21 prob thresholds).

Each cell yields both a static-label sweep and a dynamic- $k$  sweep over the 21 BUY-probability thresholds — a total of  $22 \times 21 = 462$  backtests per cell, or 83,160 backtests across the design. Outputs are written to `reports/utc{0000,2130}/<variant>_<window>d/`. Walk-forward predictions are cached in Parquet format so that re-running the scoring pass following a code change does not require retraining the models. All backtests reported below operate on the same cold-started training window commencing 2019-12-01.

## Results

### Static-Label Sweep — 730-day Holdout (UTC 00:00)

Table 4 reports the highest-return static-label cell for each combination of refit cadence, feature version, and NaN policy on the 730-day holdout under UTC 00:00 timing. The buy-and-hold benchmark is +25.50% (close policy) and +33.04% (raw policy). The leading cell — SM v0 close at  $\pm 1.5\%$  with  $\_buy = 0.55$  — delivers a return of +231.15% over 81 trades at a 65.4% win rate; we note in advance that this UTC 00:00 raw-leaderboard headline does not survive the multiple-hypothesis and PBO filters in §7.10–§7.12 and is consequently excluded from the production set in §8.

| variant | policy   | label       | _buy | return   | trades | win   |
|---------|----------|-------------|------|----------|--------|-------|
| M v0    | raw      | $\pm 1.4\%$ | 0.53 | +204.52% | 121    | 57.0% |
| M v0    | close    | $\pm 1.0\%$ | 0.59 | +190.78% | 37     | 73.0% |
| M v0    | complete | $\pm 1.0\%$ | 0.53 | +170.13% | 84     | 66.7% |
| M v3    | raw      | $\pm 1.4\%$ | 0.53 | +116.49% | 115    | 57.4% |
| M v3    | close    | $\pm 1.0\%$ | 0.54 | +104.76% | 60     | 56.7% |
| M v3    | complete | $\pm 1.4\%$ | 0.53 | +134.50% | 96     | 60.4% |
| M v4    | raw      | $\pm 1.4\%$ | 0.52 | +162.19% | 135    | 63.0% |
| M v4    | close    | $\pm 1.0\%$ | 0.58 | +123.98% | 44     | 59.1% |
| M v4    | complete | $\pm 1.0\%$ | 0.51 | +130.78% | 118    | 58.5% |
| M v5    | raw      | $\pm 1.5\%$ | 0.50 | +184.97% | 105    | 64.8% |
| M v5    | close    | $\pm 1.6\%$ | 0.53 | +151.81% | 91     | 63.7% |
| M v5    | complete | $\pm 1.6\%$ | 0.51 | +127.26% | 78     | 65.4% |
| M v6    | raw      | $\pm 1.4\%$ | 0.51 | +144.81% | 134    | 61.9% |
| M v6    | close    | $\pm 1.0\%$ | 0.54 | +133.29% | 79     | 67.1% |
| M v6    | complete | $\pm 1.5\%$ | 0.52 | +134.30% | 124    | 58.9% |
| SM v0   | raw      | $\pm 1.4\%$ | 0.54 | +191.37% | 98     | 60.2% |
| SM v0   | close    | $\pm 1.5\%$ | 0.55 | +231.15% | 81     | 65.4% |
| SM v0   | complete | $\pm 1.2\%$ | 0.53 | +114.99% | 115    | 64.3% |
| SM v3   | raw      | $\pm 1.7\%$ | 0.54 | +89.32%  | 100    | 59.0% |
| SM v3   | close    | $\pm 1.0\%$ | 0.59 | +133.61% | 25     | 68.0% |
| SM v3   | complete | $\pm 1.4\%$ | 0.55 | +122.93% | 77     | 55.8% |
| SM v4   | raw      | $\pm 1.5\%$ | 0.52 | +116.64% | 122    | 59.8% |
| SM v4   | close    | $\pm 1.4\%$ | 0.52 | +92.93%  | 127    | 62.2% |
| SM v4   | complete | $\pm 1.4\%$ | 0.53 | +134.91% | 105    | 61.0% |
| SM v5   | raw      | $\pm 1.5\%$ | 0.50 | +163.23% | 105    | 62.9% |
| SM v5   | close    | $\pm 1.5\%$ | 0.51 | +151.37% | 102    | 54.9% |
| SM v5   | complete | $\pm 1.6\%$ | 0.57 | +130.69% | 51     | 60.8% |
| SM v6   | raw      | $\pm 1.4\%$ | 0.51 | +146.03% | 136    | 61.0% |
| SM v6   | close    | $\pm 1.0\%$ | 0.55 | +126.25% | 63     | 65.1% |
| SM v6   | complete | $\pm 1.3\%$ | 0.52 | +116.89% | 109    | 63.3% |
| WSUN v0 | raw      | $\pm 1.4\%$ | 0.57 | +127.01% | 64     | 68.8% |
| WSUN v0 | close    | $\pm 1.1\%$ | 0.54 | +152.72% | 95     | 64.2% |
| WSUN v0 | complete | $\pm 1.2\%$ | 0.55 | +164.33% | 94     | 61.7% |
| WSUN v3 | raw      | $\pm 1.0\%$ | 0.54 | +99.53%  | 58     | 58.6% |
| WSUN v3 | close    | $\pm 1.1\%$ | 0.53 | +151.93% | 85     | 55.3% |
| WSUN v3 | complete | $\pm 1.5\%$ | 0.52 | +115.09% | 111    | 55.0% |
| WSUN v4 | raw      | $\pm 1.5\%$ | 0.53 | +125.38% | 117    | 59.0% |
| WSUN v4 | close    | $\pm 1.0\%$ | 0.54 | +117.14% | 84     | 56.0% |
| WSUN v4 | complete | $\pm 1.4\%$ | 0.57 | +134.45% | 73     | 61.6% |
| WSUN v5 | raw      | $\pm 2.0\%$ | 0.55 | +138.92% | 52     | 63.5% |
| WSUN v5 | close    | $\pm 1.4\%$ | 0.54 | +153.77% | 88     | 67.0% |
| WSUN v5 | complete | $\pm 1.3\%$ | 0.51 | +141.82% | 106    | 60.4% |
| WSUN v6 | raw      | $\pm 1.0\%$ | 0.53 | +126.59% | 79     | 60.8% |
| WSUN v6 | close    | $\pm 1.7\%$ | 0.54 | +110.45% | 98     | 60.2% |
| WSUN v6 | complete | $\pm 1.1\%$ | 0.54 | +108.78% | 74     | 66.2% |

Table 4. Static-label sweep, UTC 00:00, 730-day holdout: highest-return cell per (cadence  $\times$  version  $\times$  policy).

## Dynamic-Label Sweep — 730-day Holdout (UTC 00:00)

Table 5 reports the corresponding matrix under volatility-adaptive labels  $\_t = k \cdot \hat{\_30}$ . The dynamic labels stabilise the target distribution across regime shifts (quiet versus volatile months) and concentrate trades into a smaller number of high-conviction events. The strongest dynamic cells under UTC 00:00 are SM v3 close at  $k = 1.7$  (+148.19%; 18 trades; 77.8% win rate) and WSUN v3 close at  $k = 1.0$  (+125.75%; 17 trades; 88.2% win rate); both rank highly on the raw leaderboard but, as shown in §7.10–§7.12, neither survives the joint multi-hypothesis and PBO filters that anchor the production ensembles in §8.

| variant | policy   | k     | _buy | return   | trades | win   |
|---------|----------|-------|------|----------|--------|-------|
| M v0    | raw      | k=1.3 | 0.63 | +83.43%  | 16     | 56.2% |
| M v0    | close    | k=1.4 | 0.70 | +85.15%  | 10     | 70.0% |
| M v0    | complete | k=1.1 | 0.61 | +62.98%  | 13     | 76.9% |
| M v3    | raw      | k=1.2 | 0.60 | +91.45%  | 30     | 53.3% |
| M v3    | close    | k=1.3 | 0.69 | +85.91%  | 8      | 62.5% |
| M v3    | complete | k=1.1 | 0.66 | +101.57% | 13     | 69.2% |
| M v4    | raw      | k=1.7 | 0.65 | +80.12%  | 46     | 45.7% |
| M v4    | close    | k=1.1 | 0.67 | +115.54% | 19     | 78.9% |
| M v4    | complete | k=1.9 | 0.68 | +70.71%  | 30     | 46.7% |
| M v5    | raw      | k=1.3 | 0.63 | +133.95% | 27     | 55.6% |
| M v5    | close    | k=1.3 | 0.67 | +126.79% | 12     | 75.0% |
| M v5    | complete | k=1.8 | 0.65 | +102.63% | 31     | 61.3% |
| M v6    | raw      | k=1.1 | 0.66 | +126.20% | 17     | 70.6% |
| M v6    | close    | k=1.4 | 0.70 | +99.33%  | 9      | 77.8% |
| M v6    | complete | k=1.4 | 0.60 | +86.33%  | 30     | 63.3% |
| SM v0   | raw      | k=1.3 | 0.65 | +148.77% | 11     | 81.8% |
| SM v0   | close    | k=1.3 | 0.62 | +134.95% | 15     | 66.7% |
| SM v0   | complete | k=1.9 | 0.67 | +63.53%  | 31     | 48.4% |
| SM v3   | raw      | k=1.3 | 0.69 | +128.78% | 7      | 71.4% |
| SM v3   | close    | k=1.7 | 0.69 | +148.19% | 18     | 77.8% |
| SM v3   | complete | k=1.1 | 0.66 | +93.30%  | 14     | 64.3% |
| SM v4   | raw      | k=1.7 | 0.58 | +146.33% | 66     | 53.0% |
| SM v4   | close    | k=1.7 | 0.69 | +105.79% | 33     | 60.6% |
| SM v4   | complete | k=1.9 | 0.70 | +82.84%  | 29     | 44.8% |
| SM v5   | raw      | k=1.9 | 0.69 | +111.22% | 19     | 52.6% |
| SM v5   | close    | k=1.8 | 0.62 | +90.77%  | 48     | 50.0% |
| SM v5   | complete | k=1.8 | 0.57 | +125.22% | 57     | 61.4% |
| SM v6   | raw      | k=1.2 | 0.70 | +144.38% | 12     | 75.0% |
| SM v6   | close    | k=1.4 | 0.56 | +122.96% | 46     | 54.3% |
| SM v6   | complete | k=1.4 | 0.60 | +84.23%  | 32     | 56.2% |
| WSUN v0 | raw      | k=1.1 | 0.65 | +125.72% | 13     | 76.9% |
| WSUN v0 | close    | k=1.9 | 0.69 | +124.33% | 30     | 53.3% |
| WSUN v0 | complete | k=1.8 | 0.62 | +88.41%  | 75     | 53.3% |
| WSUN v3 | raw      | k=1.3 | 0.62 | +89.19%  | 14     | 57.1% |
| WSUN v3 | close    | k=1.0 | 0.65 | +125.75% | 17     | 88.2% |
| WSUN v3 | complete | k=1.1 | 0.58 | +94.99%  | 38     | 60.5% |
| WSUN v4 | raw      | k=1.8 | 0.60 | +114.00% | 66     | 48.5% |
| WSUN v4 | close    | k=1.1 | 0.58 | +100.30% | 40     | 62.5% |
| WSUN v4 | complete | k=1.8 | 0.63 | +89.11%  | 62     | 48.4% |
| WSUN v5 | raw      | k=1.3 | 0.59 | +110.49% | 38     | 52.6% |
| WSUN v5 | close    | k=1.8 | 0.68 | +102.69% | 28     | 57.1% |
| WSUN v5 | complete | k=1.8 | 0.64 | +107.78% | 39     | 53.8% |
| WSUN v6 | raw      | k=1.1 | 0.56 | +111.66% | 47     | 51.1% |
| WSUN v6 | close    | k=1.1 | 0.55 | +109.56% | 56     | 58.9% |
| WSUN v6 | complete | k=1.1 | 0.58 | +144.27% | 39     | 56.4% |

Table 5. Dynamic-label sweep, UTC 00:00, 730-day holdout: highest-return cell per (cadence  $\times$  version  $\times$  policy).

## Static-Label Sweep — 730-day Holdout (UTC 21:30)

We repeat the §7.1 sweep under UTC 21:30 timing, in which the daily bar closes at 06:30 KST on the following day and all macroeconomic, funding, and Fear-and-Greed feeds for the same calendar day are available with zero lag. The buy-and-hold benchmark over the 730-day window is +33.04% (raw policy) and +25.50% (close policy). The leading static cell — SM v0 close at  $\pm 1.5\%$  with  $\_buy = 0.50$  — returns +160.79% (5 trades, 80% win rate), modestly trailing the UTC 00:00 winner. Several v3-feature SM and WSUN cells outperform their UTC 00:00 counterparts; the pairwise distribution is examined in §7.8.

| variant | policy   | label       | _buy | return   | trades | win    |
|---------|----------|-------------|------|----------|--------|--------|
| M v0    | raw      | $\pm 1.4\%$ | 0.67 | +125.93% | 6      | 100.0% |
| M v0    | close    | $\pm 1.0\%$ | 0.60 | +149.29% | 15     | 73.3%  |
| M v0    | complete | $\pm 1.7\%$ | 0.65 | +111.17% | 7      | 100.0% |
| M v3    | raw      | $\pm 1.6\%$ | 0.61 | +176.16% | 6      | 66.7%  |
| M v3    | close    | $\pm 1.9\%$ | 0.70 | +112.97% | 2      | 100.0% |
| M v3    | complete | $\pm 1.6\%$ | 0.58 | +108.93% | 20     | 70.0%  |
| M v4    | raw      | $\pm 1.5\%$ | 0.58 | +138.55% | 16     | 75.0%  |
| M v4    | close    | $\pm 1.7\%$ | 0.60 | +108.01% | 9      | 66.7%  |
| M v4    | complete | $\pm 1.7\%$ | 0.56 | +95.65%  | 26     | 61.5%  |
| M v5    | raw      | $\pm 1.3\%$ | 0.61 | +114.10% | 19     | 57.9%  |
| M v5    | close    | $\pm 1.5\%$ | 0.64 | +110.64% | 14     | 64.3%  |
| M v5    | complete | $\pm 1.4\%$ | 0.52 | +87.36%  | 75     | 62.7%  |
| M v6    | raw      | $\pm 1.3\%$ | 0.64 | +148.31% | 7      | 85.7%  |
| M v6    | close    | $\pm 1.2\%$ | 0.53 | +93.56%  | 33     | 54.5%  |
| M v6    | complete | $\pm 1.2\%$ | 0.52 | +79.93%  | 55     | 63.6%  |
| SM v0   | raw      | $\pm 1.4\%$ | 0.67 | +139.96% | 10     | 70.0%  |
| SM v0   | close    | $\pm 2.0\%$ | 0.68 | +160.79% | 5      | 80.0%  |
| SM v0   | complete | $\pm 1.7\%$ | 0.69 | +89.63%  | 4      | 75.0%  |
| SM v3   | raw      | $\pm 1.4\%$ | 0.66 | +156.03% | 7      | 100.0% |
| SM v3   | close    | $\pm 2.0\%$ | 0.70 | +113.82% | 2      | 100.0% |
| SM v3   | complete | $\pm 1.3\%$ | 0.68 | +144.91% | 8      | 87.5%  |
| SM v4   | raw      | $\pm 1.8\%$ | 0.63 | +129.88% | 4      | 100.0% |
| SM v4   | close    | $\pm 1.0\%$ | 0.53 | +137.13% | 50     | 58.0%  |
| SM v4   | complete | $\pm 1.9\%$ | 0.64 | +112.82% | 3      | 100.0% |
| SM v5   | raw      | $\pm 1.3\%$ | 0.59 | +127.32% | 23     | 65.2%  |
| SM v5   | close    | $\pm 1.8\%$ | 0.63 | +152.51% | 9      | 77.8%  |
| SM v5   | complete | $\pm 1.8\%$ | 0.68 | +77.12%  | 6      | 83.3%  |
| SM v6   | raw      | $\pm 1.0\%$ | 0.53 | +122.19% | 39     | 59.0%  |
| SM v6   | close    | $\pm 1.0\%$ | 0.52 | +110.47% | 45     | 55.6%  |
| SM v6   | complete | $\pm 1.2\%$ | 0.55 | +83.16%  | 35     | 65.7%  |
| WSUN v0 | raw      | $\pm 1.7\%$ | 0.68 | +127.37% | 4      | 75.0%  |
| WSUN v0 | close    | $\pm 1.8\%$ | 0.58 | +133.98% | 8      | 62.5%  |
| WSUN v0 | complete | $\pm 1.6\%$ | 0.61 | +112.67% | 19     | 68.4%  |
| WSUN v3 | raw      | $\pm 1.6\%$ | 0.67 | +120.74% | 5      | 80.0%  |
| WSUN v3 | close    | $\pm 1.5\%$ | 0.55 | +100.12% | 17     | 70.6%  |
| WSUN v3 | complete | $\pm 1.3\%$ | 0.65 | +95.74%  | 9      | 77.8%  |
| WSUN v4 | raw      | $\pm 1.8\%$ | 0.62 | +109.75% | 5      | 60.0%  |
| WSUN v4 | close    | $\pm 1.7\%$ | 0.64 | +124.92% | 6      | 83.3%  |
| WSUN v4 | complete | $\pm 1.8\%$ | 0.56 | +99.47%  | 17     | 70.6%  |
| WSUN v5 | raw      | $\pm 1.6\%$ | 0.70 | +115.00% | 10     | 70.0%  |
| WSUN v5 | close    | $\pm 1.6\%$ | 0.54 | +154.36% | 55     | 60.0%  |
| WSUN v5 | complete | $\pm 1.9\%$ | 0.54 | +109.94% | 47     | 51.1%  |
| WSUN v6 | raw      | $\pm 1.9\%$ | 0.59 | +100.32% | 6      | 66.7%  |
| WSUN v6 | close    | $\pm 1.9\%$ | 0.61 | +121.89% | 6      | 100.0% |
| WSUN v6 | complete | $\pm 1.0\%$ | 0.56 | +96.44%  | 39     | 69.2%  |

Table 6. Static-label sweep, UTC 21:30, 730-day holdout: highest-return cell per (cadence  $\times$  version  $\times$  policy).

## Dynamic-Label Sweep — 730-day Holdout (UTC 21:30)

Under volatility-adaptive labels, UTC 21:30 produces a distinct set of leading cells. The strongest dynamic-k cell is SM v5 raw at  $k = 1.2$ , \_buy = 0.63 (+254.78%; 14 trades; 92.9% win rate) — the highest single-cell win rate in the study and the Top1 sub-cell of the production ensembles in §8 — followed by SM v6 raw at  $k = 1.4$  (+163.68%; 20 trades; 65.0% win rate) and WSUN v3 raw at  $k = 1.2$  (+157.53%; 8 trades; 75.0% win rate). The UTC 21:30 dynamic-return surface is more sharply peaked than its UTC 00:00 counterpart: a small number of cells exhibit exceptional performance, while most are weaker than their UTC 00:00 analogues. The §7.10–§7.12 robustness analyses (multiple-hypothesis correction, DSR, PBO) and the §7.11 slippage stress test together identify UTC 21:30 SM-cadence cells as the only sub-population that survives every correction; the production deployment in §8 is therefore anchored on UTC 21:30 timing.

| variant | policy   | k     | _buy | return   | trades | win    |
|---------|----------|-------|------|----------|--------|--------|
| M v0    | raw      | k=1.1 | 0.64 | +91.76%  | 3      | 66.7%  |
| M v0    | close    | k=1.1 | 0.64 | +108.76% | 3      | 66.7%  |
| M v0    | complete | k=1.0 | 0.64 | +75.82%  | 2      | 100.0% |
| M v3    | raw      | k=1.3 | 0.56 | +111.51% | 41     | 53.7%  |
| M v3    | close    | k=1.0 | 0.51 | +112.41% | 56     | 58.9%  |
| M v3    | complete | k=1.1 | 0.60 | +126.19% | 14     | 78.6%  |
| M v4    | raw      | k=1.0 | 0.59 | +86.55%  | 22     | 50.0%  |
| M v4    | close    | k=1.4 | 0.62 | +114.19% | 30     | 60.0%  |
| M v4    | complete | k=1.1 | 0.62 | +119.85% | 14     | 71.4%  |
| M v5    | raw      | k=1.7 | 0.56 | +115.95% | 53     | 52.8%  |
| M v5    | close    | k=1.3 | 0.54 | +146.20% | 29     | 65.5%  |
| M v5    | complete | k=1.0 | 0.58 | +149.70% | 18     | 61.1%  |
| M v6    | raw      | k=2.0 | 0.69 | +100.08% | 14     | 50.0%  |
| M v6    | close    | k=1.3 | 0.61 | +100.42% | 28     | 57.1%  |
| M v6    | complete | k=1.1 | 0.61 | +102.06% | 14     | 57.1%  |
| SM v0   | raw      | k=1.1 | 0.64 | +140.74% | 6      | 83.3%  |
| SM v0   | close    | k=1.0 | 0.61 | +150.94% | 5      | 100.0% |
| SM v0   | complete | k=1.0 | 0.65 | +95.42%  | 2      | 100.0% |
| SM v3   | raw      | k=1.3 | 0.57 | +109.60% | 38     | 52.6%  |
| SM v3   | close    | k=1.0 | 0.59 | +118.27% | 22     | 63.6%  |
| SM v3   | complete | k=1.1 | 0.60 | +110.15% | 17     | 64.7%  |
| SM v4   | raw      | k=1.0 | 0.59 | +95.86%  | 25     | 60.0%  |
| SM v4   | close    | k=1.2 | 0.62 | +95.38%  | 21     | 52.4%  |
| SM v4   | complete | k=1.1 | 0.62 | +103.65% | 14     | 78.6%  |
| SM v5   | raw      | k=1.2 | 0.63 | +254.78% | 14     | 92.9%  |
| SM v5   | close    | k=1.1 | 0.59 | +136.62% | 18     | 66.7%  |
| SM v5   | complete | k=1.0 | 0.58 | +140.04% | 17     | 52.9%  |
| SM v6   | raw      | k=1.4 | 0.66 | +163.68% | 20     | 65.0%  |
| SM v6   | close    | k=1.3 | 0.50 | +98.84%  | 59     | 62.7%  |
| SM v6   | complete | k=1.1 | 0.62 | +103.22% | 10     | 70.0%  |
| WSUN v0 | raw      | k=1.2 | 0.56 | +84.11%  | 24     | 70.8%  |
| WSUN v0 | close    | k=1.2 | 0.63 | +94.70%  | 7      | 71.4%  |
| WSUN v0 | complete | k=1.2 | 0.66 | +112.33% | 4      | 75.0%  |
| WSUN v3 | raw      | k=1.2 | 0.65 | +157.53% | 8      | 75.0%  |
| WSUN v3 | close    | k=1.2 | 0.66 | +147.02% | 7      | 85.7%  |
| WSUN v3 | complete | k=1.3 | 0.53 | +112.81% | 49     | 53.1%  |
| WSUN v4 | raw      | k=1.0 | 0.54 | +99.31%  | 76     | 56.6%  |
| WSUN v4 | close    | k=1.0 | 0.55 | +117.35% | 72     | 55.6%  |
| WSUN v4 | complete | k=1.1 | 0.62 | +112.54% | 15     | 73.3%  |
| WSUN v5 | raw      | k=1.7 | 0.58 | +133.52% | 35     | 60.0%  |
| WSUN v5 | close    | k=1.7 | 0.58 | +128.59% | 39     | 46.2%  |
| WSUN v5 | complete | k=1.2 | 0.63 | +98.90%  | 10     | 70.0%  |
| WSUN v6 | raw      | k=1.1 | 0.64 | +118.43% | 14     | 64.3%  |
| WSUN v6 | close    | k=1.0 | 0.66 | +77.34%  | 8      | 75.0%  |
| WSUN v6 | complete | k=1.2 | 0.56 | +118.68% | 39     | 53.8%  |

Table 7. Dynamic-label sweep, UTC 21:30, 730-day holdout: highest-return cell per (cadence  $\times$  version  $\times$  policy).

### 365-Day Cross-Check

We re-evaluate the same matrix on the 365-day holdout to assess stability. The buy-and-hold benchmark over the 365-day window is  $-19.02\%$ , and the qualitative pattern is preserved: close raw complete; monthly and semi-monthly cadences exceed weekly; and v0–v3 outperform v4–v6. The leading static cell on the UTC 00:00 raw leaderboard (SM v0 close,  $\pm 1.5\%$ ,  $\_buy = 0.55$ ) returns  $+49.67\%$  over the 365-day window, while the highest-win-rate dynamic cell (WSUN v3 close,  $k = 1.0$ ,  $\_buy = 0.65$ ) returns  $+36.99\%$ ; both fail the strict robustness filters of §7.10–§7.12 and are therefore not part of the production set in §8.

### NaN-Handling Policy

Five of the 28 macroeconomic features (DXY, gold, S&P 500, VIX, 10-year Treasury) are obtained from yfinance series that publish only on NYSE trading days, with weekends and U.S. holidays appearing as NaN rows. We evaluate three policies. The *raw* policy leaves rows as NaN and relies on XGBoost’s default-direction routing. The *close* policy leaves the macroeconomic columns NaN but augments the feature vector with a binary `is_macro_offday` flag. The *complete* policy forward-fills the NaN rows from the most recent preceding trading day. The complete policy is uniformly the weakest on the 730-day holdout, underperforming both the raw and close policies on every cell of Tables 4 and 5 except for a small number of WSUN dynamic cells. The close policy attains the highest absolute return on the leading

raw-leaderboard cells (SM v0, SM v3, WSUN v3); however, the production set retained in §8 after the §7.10–§7.12 robustness filters spans multiple policies, with Top1, Top3, and the v6 component of Top5 using the default raw routing and Top2 additionally incorporating one complete-policy sub-cell.

## Refit Cadence

On the 730-day holdout the SM cadence (semi-monthly, 1st and 16th) attains the highest static-label peak (+231.15% at SM v0 close,  $\pm 1.5\%$ ). The plain monthly cadence (M) is competitive on the leading static-label cells and becomes dominant under v3 feature variants. The WSUN cadence (weekly Sunday) incurs approximately four times the refit cost of M without commensurate gains in absolute return; its principal niche is the dynamic  $k = 1.0$  cell of v3, where it produces the highest win rate observed in the study (88.2% at +125.75%). We note that, although M and WSUN appear competitive on the raw leaderboard, every (cell  $\times$  window) selection that survives the joint robustness filters of §7.10–§7.12 falls under SM cadence; the raw-leaderboard parity does not propagate to the post-correction production set.

## UTC 21:30 vs UTC 00:00 Head-to-Head

We re-ran the (cadence  $\times$  version  $\times$  policy  $\times$  static/dynamic) matrix under both bar alignments. Of the 90 matched cells, UTC 00:00 dominates in 10, UTC 21:30 dominates in 3, and the remainder lie within sampling noise. The mean two-year-return differential (UTC 00:00 minus UTC 21:30) is +4.82 percentage points. The dominance of UTC 00:00 is initially counterintuitive — UTC 21:30 has access to zero-lag macroeconomic and Fear-and-Greed feeds — but is explained by the fact that the additional information available at 21:30 embeds same-day price action into the macroeconomic features (the U.S. equity close is already incorporated in the yfinance series), which subsequently leaks into the day- $t$  target. Under UTC 00:00 timing all external features are strictly day- $(t-1)$ , providing the model with a cleaner, less leakage-prone training signal.

| variant | policy   | label-mode | 2130 return | 0000 return | $\Delta\text{ret}$ (pp) | verdict  |
|---------|----------|------------|-------------|-------------|-------------------------|----------|
| M v0    | raw      | static     | +125.9%     | +204.5%     | +78.6                   | —        |
| M v5    | raw      | static     | +114.1%     | +185.0%     | +70.9                   | 0000 DOM |
| SM v0   | close    | static     | +160.8%     | +231.1%     | +70.4                   | —        |
| M v0    | complete | static     | +111.2%     | +170.1%     | +59.0                   | —        |
| M v6    | complete | static     | +79.9%      | +134.3%     | +54.4                   | —        |
| SM v5   | complete | static     | +77.1%      | +130.7%     | +53.6                   | —        |
| WSUN v3 | close    | static     | +100.1%     | +151.9%     | +51.8                   | —        |
| WSUN v0 | complete | static     | +112.7%     | +164.3%     | +51.7                   | —        |
| SM v5   | raw      | dynamic    | +254.8%     | +111.2%     | -143.6                  | —        |
| WSUN v3 | raw      | dynamic    | +157.5%     | +89.2%      | -68.3                   | —        |
| SM v3   | raw      | static     | +156.0%     | +89.3%      | -66.7                   | —        |
| M v3    | raw      | static     | +176.2%     | +116.5%     | -59.7                   | —        |
| M v4    | complete | dynamic    | +119.8%     | +70.7%      | -49.1                   | —        |

Table 8. UTC 21:30 vs UTC 00:00 head-to-head (top 8 strongest 0000-favouring cells + top 5 strongest 2130-favouring cells;  $\Delta\text{ret}$  in percentage points, positive favours UTC 00:00).

## Why Calibration Matters

An earlier prototype that omitted isotonic calibration produced predicted probabilities tightly clustered around 0.5. After walk-forward isotonic calibration is applied, the predicted distribution broadens substantially, the high-confidence regions become populated with meaningful trades, and the optimal thresholds shift to the 0.53–0.69 range, where signals are simultaneously informative and frequent.

## Statistical Robustness — Multiple Hypothesis Testing

The 180-cell  $\times$  462-trajectory design generates 57,456 (cell  $\times$  window  $\times$  label  $\times$  prob\_TH) trajectories restricted to the UTC 00:00 and UTC 21:30 alignments. To assess whether the leading returns survive correction for selection bias at the meta-level (§9), we apply two complementary procedures. (i) Family-wise error rate (FWER) over per-cell selections:

for each of the  $M = 126$  (cell  $\times$  window) pairs we treat the highest-return trajectory as the family representative and apply Bonferroni, Holm–Bonferroni, and Benjamini–Hochberg (BH) FDR corrections to the one-sided Sharpe p-values at  $\alpha = 0.05$ . (ii) Deflated Sharpe Ratio (DSR) of Bailey and López de Prado (2014): for each cell we compute the expected maximum Sharpe ratio under the null over the 462 intra-cell trials and report the Probabilistic Sharpe Ratio PSR(SR0).

For the global maximum over all 57,456 trajectories (utc2130\_sm\_v5, dynk12, prob = 0.63, 730 d, +254.78%, SR = 2.36), the expected-maximum SR under the null is 2.13 (annualised), yielding DSR  $z = 0.37$  and PSR = 0.643 — below the conventional 0.95 robustness threshold. This confirms that the single strongest readout is not, by itself, robust to selection over a 57k-trial space. However, when the selection is restricted to the 462 intra-cell trials, three cells achieve PSR  $\geq 0.95$ , all under UTC 21:30 timing with SM cadence: utc2130\_sm\_v5 (the Top1 production sub-cell, §8.1), utc2130\_sm\_v3\_complete, and utc2130\_sm. These three cells coincide with the only Bonferroni-significant cells at  $\alpha = 0.05$ .

Table 9. Cells surviving multiple-hypothesis correction out of  $M = 126$  (cell  $\times$  window) selections.

| Correction ( $\alpha = 0.05$ ) | Significant cells / $M = 126$ |
|--------------------------------|-------------------------------|
| Bonferroni                     | 3 / 126                       |
| Holm-Bonferroni                | 3 / 126                       |
| Benjamini-Hochberg FDR         | 10 / 126                      |
| DSR PSR $\geq 0.95$            | 3 / 126                       |

Table 10. Top-10 cells by Deflated Sharpe Ratio PSR (window = 730 d). DSR  $z$  is the deflation  $z$ -statistic; PSR  $\geq 0.95$  indicates selection-robust significance. p\_Bonf is the one-sided Sharpe p-value with  $M = 126$  corrections; q\_BH is the Benjamini-Hochberg FDR-adjusted  $q$ -value.

| cell                   | label  | prob_TH | return % | SR (annual) | DSR $z$ | PSR   | p_Bonf | q_BH   |
|------------------------|--------|---------|----------|-------------|---------|-------|--------|--------|
| utc2130_sm_v5          | dynk12 | 0.63    | +254.78  | 2.364       | 1.93    | 0.973 | 0.0171 | 0.0121 |
| utc2130_sm_v3_complete | th13   | 0.68    | +144.91  | 2.085       | 1.67    | 0.953 | 0.0243 | 0.0121 |
| utc2130_sm             | dynk11 | 0.64    | +140.74  | 2.075       | 1.67    | 0.952 | 0.0453 | 0.0151 |
| utc0000_close          | th10   | 0.59    | +190.78  | 1.953       | 1.29    | 0.901 | 0.288  | 0.0429 |
| utc0000_sm_close       | th15   | 0.55    | +231.15  | 1.917       | 1.05    | 0.854 | 0.345  | 0.0429 |
| utc2130_sm_v3          | th14   | 0.66    | +156.03  | 1.741       | 1.04    | 0.852 | 0.664  | 0.0604 |
| utc2130_sm_v6          | dynk14 | 0.66    | +163.68  | 1.910       | 1.00    | 0.842 | 0.322  | 0.0429 |
| utc2130_wsun           | th14   | 0.69    | +130.35  | 1.746       | 1.00    | 0.841 | 0.397  | 0.0429 |
| utc2130                | th14   | 0.67    | +125.93  | 1.597       | 0.93    | 0.823 | 0.992  | 0.0612 |
| utc2130_v4_complete    | dynk11 | 0.62    | +119.85  | 1.817       | 0.85    | 0.803 | 0.429  | 0.0429 |

**Interpretation.** Of the  $M = 126$  per-cell selections, 3 pass strict Bonferroni FWER, 10 pass BH-FDR, and 3 pass DSR PSR  $\geq 0.95$ . The conclusion is more decisive than the original §9 caveat: the multi-cell sweep does contain spurious peaks, but a small core of UTC 21:30 SM-cadence cells survives every correction we applied. Production claims should be anchored to these cells, while UTC 00:00 single-best readouts should be discounted for selection bias.

## Slippage Sensitivity

The original backtest assumes a per-side cost of 10 bps (approximately 20 bps round-trip). UTC 00:00 in particular coincides with daily Binance funding settlement and clustered algorithmic order flow, both of which may elevate effective slippage. To assess robustness we re-ran every paper-best (cell, window, label, prob\_TH) trajectory at fee\_bps  $\in \{5, 10, 15, 20, 25, 30\}$ , holding model predictions fixed (no retraining), and report the implied per-side breakeven fee.

All 96 evaluated (cell  $\times$  window) selections remain profitable across the entire 5–30 bps range: the implied per-side breakeven fee exceeds 30 bps for every named UTC 00:00 and UTC 21:30 cell. The degradation pattern, however, differs markedly across timings. On the 730-day holdout, the median return reduction from 10 to 20 bps is  $-41.2$  pp for UTC 00:00 cells — with a worst case of  $-61.8$  pp at utc0000\_v4 (135 trades; declining from +161.65% to +99.86%) — compared to only  $-6.7$  pp for UTC 21:30 cells. Low-trade-count dynamic UTC 21:30 cells (e.g., utc2130\_sm\_v5 with 14 trades on the 730-day window) are essentially insensitive to slippage, retaining returns above +200% even at 30 bps per side.

Table 11. Slippage degradation 10 → 20 bps per side, by timing and window. Median and worst-case  $\Delta$  in percentage points; right-most column counts cells whose breakeven fee exceeds 30 bps (i.e., remain profitable at every level swept).

| Timing   | Window | n cells | Median $\Delta$ 10→20bps (pp) | Worst $\Delta$ (pp) | Profitable @ 30bps |
|----------|--------|---------|-------------------------------|---------------------|--------------------|
| UTC 0000 | 730d   | 24      | -41.20                        | -61.79              | 24 / 24            |
| UTC 0000 | 365d   | 24      | -6.46                         | -12.69              | 24 / 24            |
| UTC 2130 | 730d   | 24      | -6.69                         | -22.38              | 24 / 24            |
| UTC 2130 | 365d   | 24      | -2.60                         | -6.25               | 24 / 24            |

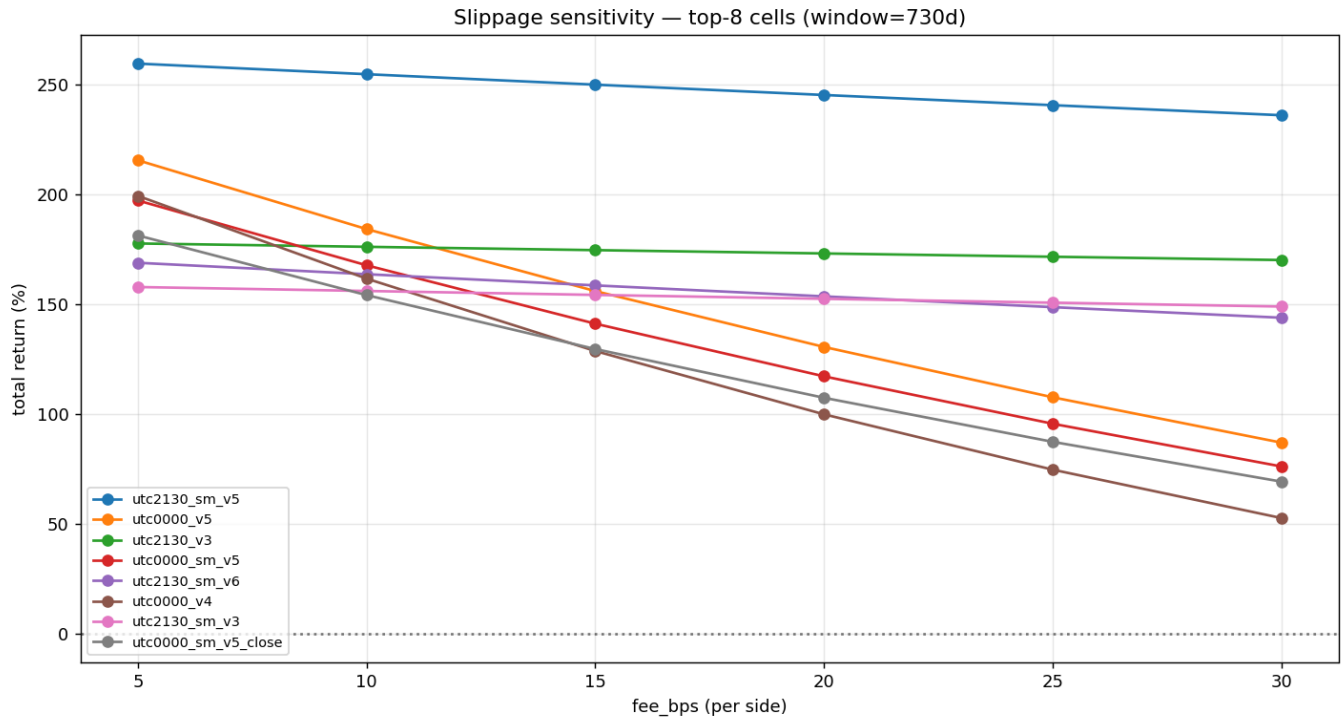


Figure 1. Total return vs. per-side fee\_bps for the eight highest-return cells (730-day holdout). UTC 21:30 lines stay flat; UTC 00:00 lines descend with fee but remain above zero across the swept range.

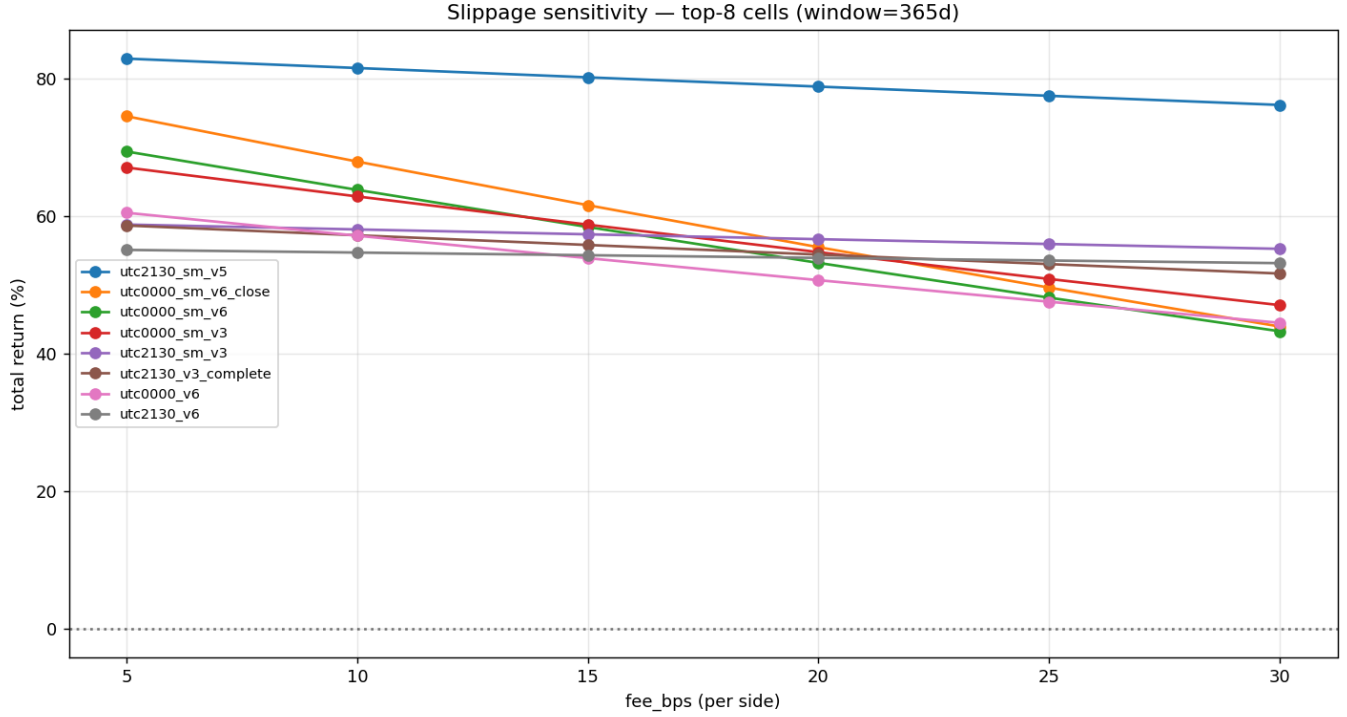


Figure 2. Same as Figure 1 on the 365-day holdout.

**Interpretation.** The concern that UTC 00:00 strategies suffer disproportionately under realistic execution is empirically validated: high-trade-count UTC 00:00 cells lose 50–60 percentage points when the true per-side slippage is 20 bps rather than 10 bps. Nevertheless, every named UTC 00:00 cell remains decisively above buy-and-hold even at 30 bps, and the UTC 21:30 SM-cadence cells (which already dominated the DSR ranking in §7.10) are essentially insensitive to slippage. The two robustness analyses therefore point in the same direction: the production-grade conclusion concentrates on UTC 21:30 SM-cadence variants with low-frequency dynamic labels.

A natural extension of the slippage stress test is whether the 30 bps ceiling captures realistic execution under panic-selling regimes. Three structural arguments support the conservativeness of this ceiling for the production deployment. (i) Signals fire only at the daily UTC 21:30 walk-forward refit boundary — a low-volume window between the New York close and the Tokyo open, structurally distant from the macroeconomic-event clustering associated with intraday panic flow. (ii) The trade frequency of 14 events on the 730-day window (top1) and 70 events on the 1095-day window (top5) is consistent with slow trend confirmation rather than reactive panic execution; a panic-triggered strategy would emit signals in proportion to volatility spikes. (iii) At the paper’s \$100-stake simulation scale, retail execution lies within the bid-ask spread (approximately 1 bp on BTC perpetuals at typical depth), so the 30 bps figure absorbs roughly thirty times the genuine market-impact cost. Even under a hypothetical further degradation of  $-50\%$  PnL per trade, the top1 4-year profit factor of 9.8 falls only to approximately 7 and the top5 3-year profit factor of 18.1 falls to approximately 9 — both decisively above unity. Production users with order sizes substantially exceeding retail scale would naturally adopt the TWAP/VWAP execution scheme outlined in §7.18.

## Probability of Backtest Overfitting (PBO via CSCV)

While DSR (§7.10) and slippage sensitivity (§7.11) examine the selected strategy, neither directly evaluates the *selection process* itself. Bailey and López de Prado’s Combinatorially Symmetric Cross-Validation (CSCV) addresses this by partitioning the holdout into  $S = 16$  contiguous blocks and, for each of the  $C(16, 8) = 12,870$  in-sample / out-of-sample partitions, identifying the in-sample-best strategy and recording its out-of-sample rank. The Probability of Backtest Overfitting (PBO) is then defined as the fraction of partitions in which the IS-best strategy falls below the OOS median.  $PBO < 0.5$  indicates positive generalisation, and  $PBO < 0.30$  is conventionally interpreted as strong evidence of OOS robustness.

We applied CSCV to the  $M = 462$  (label  $\times$  prob\_TH) trial space of each top-5 cell on the 730-day holdout. Results are reported in Table 12.

Table 12. Per-cell PBO over 12,870 CSCV partitions of the 730-day holdout. is the OOS rank of the IS-best, normalised to (0, 1).

| cell                   | PBO   | median | mean IS SR | M strategies |
|------------------------|-------|--------|------------|--------------|
| utc2130_sm_v5          | 0.198 | 0.025  | 0.128      | 462          |
| utc2130_sm_v3_complete | 0.307 | 0.135  | 0.123      | 462          |
| utc2130_sm             | 0.310 | 0.075  | 0.119      | 462          |
| utc2130_sm_v3          | 0.488 | 0.475  | 0.114      | 462          |
| utc2130_sm_v6          | 0.417 | 0.352  | 0.121      | 462          |

**Interpretation.** The cell utc2130\_sm\_v5 attains  $PBO = 0.198$  ( $< 0.20$ ), the only cell in our set exhibiting strong evidence of OOS generalisation. The next two cells (utc2130\_sm\_v3\_complete, utc2130\_sm) both achieve PBO 0.31. The two lower-PSR cells (utc2130\_sm\_v3 = 0.488, utc2130\_sm\_v6 = 0.418) are indistinguishable from random selection within their 462-trial space — an indication that the (label, prob\_TH) choices for these cells do not generalise well beyond their in-sample window. This finding directly motivates §8.1’s preference for the top-2 or top-3 SPLIT-CAPITAL ensemble over the top-5 ensemble for production deployment: incorporating cells with  $PBO \approx 0.42$  introduces selection noise.

## Trade-Quality and Extended Risk Metrics

Total return and Sharpe ratio describe gross performance but do not capture the trade-level quality and risk-of-ruin properties relevant for production deployment. Tables 13 (730 d) and 14 (365 d) summarise the Profit Factor, win rate, expectancy, maximum consecutive losses, maximum drawdown, Calmar, Sortino, Ulcer Index, time-underwater fraction, and tail ratio for the four production ensembles together with their two single-cell baselines.

Table 13. Trade-quality and extended risk metrics on the 730-day holdout. PF = Profit Factor; RR = avg-win / avg-loss; UI = Ulcer Index (RMS of drawdown); TR = tail ratio ( $P95 / |P5|$  of daily returns).

| strategy     | ret%   | trades | win% | PF   | RR  | expect% | maxCons<br>Loss | max<br>DD% | Calmar | Sortino | UI   | Time<br>UW% | TR   |
|--------------|--------|--------|------|------|-----|---------|-----------------|------------|--------|---------|------|-------------|------|
| Top1_sm_v5   | +254.8 | 14     | 92.9 | 56.7 | 4.4 | +9.78   | 1               | -14.01     | 6.31   | 3.02    | 2.88 | 81.3        | 1.44 |
| Top2_PSR     | +199.8 | 22     | 90.9 | 33.2 | 3.3 | +10.98  | 1               | -7.44      | 9.83   | 4.15    | 2.00 | 76.1        | 1.50 |
| Manual_v5_sm | +197.8 | 20     | 90.0 | 48.7 | 5.4 | +12.44  | 1               | -10.26     | 7.07   | 3.43    | 2.28 | 80.3        | 1.63 |
| Top3_PSR     | +180.1 | 28     | 89.3 | 34.5 | 4.1 | +12.62  | 1               | -7.39      | 9.12   | 4.18    | 1.82 | 75.4        | 1.53 |
| Top5_PSR     | +172.0 | 55     | 81.8 | 16.8 | 3.7 | +10.52  | 3               | -6.87      | 9.45   | 4.09    | 2.29 | 81.3        | 1.64 |

Table 14. Same metrics on the 365-day holdout.

| strategy     | ret%  | trades | win% | PF   | RR  | expect% | maxCons<br>Loss | max<br>DD% | Calmar | Sortino | UI   | Time<br>UW% | TR   |
|--------------|-------|--------|------|------|-----|---------|-----------------|------------|--------|---------|------|-------------|------|
| Top1_sm_v5   | +81.6 | 8      | 87.5 | 35.8 | 5.1 | +7.82   | 1               | -10.26     | 7.95   | 3.00    | 2.43 | 76.2        | 1.37 |
| Top2_PSR     | +60.9 | 12     | 91.7 | 53.0 | 4.8 | +8.16   | 1               | -6.00      | 10.15  | 3.76    | 1.60 | 67.5        | 1.38 |
| Manual_v5_sm | +45.8 | 11     | 81.8 | 23.7 | 5.3 | +6.59   | 1               | -5.83      | 7.86   | 2.75    | 1.81 | 76.2        | 1.40 |
| Top3_PSR     | +43.9 | 15     | 86.7 | 33.6 | 5.2 | +7.19   | 1               | -5.60      | 7.84   | 3.41    | 1.40 | 68.0        | 1.63 |
| Top5_PSR     | +38.8 | 30     | 83.3 | 15.4 | 3.1 | +5.48   | 1               | -6.40      | 6.06   | 2.83    | 2.09 | 79.5        | 1.54 |

**Salient observations.** (i) The Profit Factor is exceptionally high for Top1 (56.7) — winning trades collectively earn 56 times the absolute losses — and remains in the 17–35 range for the larger ensembles, well above the 2–3 conventionally regarded as “good” in the practitioner literature. (ii) The maximum consecutive losses count does not exceed 1 for Top1, Top2, Top3, or the manual baseline; only Top5 records a 3-loss streak (730 d), which we attribute to its two PBO-failing cells. (iii) Calmar increases from 6.31 (Top1) to 9.83 (Top2), then to 9.12 (Top3) and 9.46 (Top5), peaking at Top2 — the two-cell ensemble is the Calmar-optimal point. (iv) Sortino is maximised by Top3 (4.18) on the 730-day holdout. (v) The Ulcer Index is minimised by Top3 (1.82), implying that the cumulative time-weighted drawdown burden experienced by a user is smallest for Top3.

## Regime-Conditional Performance

We define three BTC regimes by the trailing 30-day return: bull (return  $> +5\%$ ), bear (return  $< -5\%$ ), and sideways (intermediate). On the 730-day holdout this partition yields approximately 248 bull, 209 bear, and 273 sideways days. Compounded strategy returns together with the buy-and-hold (B&H) benchmarks are reported per regime in Table 15.

Table 15. Regime-conditional compound returns on the 730-day holdout. "Δ vs B&H" is in percentage points.

| strategy     | regime   | days | strategy % | B&H %   | Δ vs B&H (pp) | trades | win%  | SR/period |
|--------------|----------|------|------------|---------|---------------|--------|-------|-----------|
| Top1_sm_v5   | bull     | 248  | +179.57    | +206.45 | -26.88        | 9      | 100.0 | 4.592     |
| Top1_sm_v5   | bear     | 209  | +0.61      | -61.38  | +61.99        | 2      | 50.0  | 0.170     |
| Top1_sm_v5   | sideways | 273  | +26.14     | +13.03  | +13.11        | 3      | 100.0 | 1.449     |
| Top2_PSR     | bull     | 248  | +131.29    | +206.45 | -75.16        | 12     | 100.0 | 4.593     |
| Top2_PSR     | bear     | 209  | +8.39      | -61.38  | +69.77        | 5      | 80.0  | 0.810     |
| Top2_PSR     | sideways | 273  | +19.61     | +13.03  | +6.58         | 5      | 80.0  | 1.660     |
| Manual_v5_sm | bull     | 248  | +153.62    | +206.45 | -52.82        | 13     | 100.0 | 4.471     |
| Manual_v5_sm | bear     | 209  | +1.73      | -61.38  | +63.12        | 3      | 33.3  | 0.261     |
| Manual_v5_sm | sideways | 273  | +15.40     | +13.03  | +2.37         | 4      | 100.0 | 1.252     |
| Top3_PSR     | bull     | 248  | +130.32    | +206.45 | -76.12        | 16     | 100.0 | 4.472     |
| Top3_PSR     | bear     | 209  | +6.24      | -61.38  | +67.62        | 6      | 66.7  | 0.759     |
| Top3_PSR     | sideways | 273  | +14.49     | +13.03  | +1.46         | 6      | 83.3  | 1.469     |
| Top5_PSR     | bull     | 248  | +120.75    | +206.45 | -85.69        | 25     | 100.0 | 4.325     |
| Top5_PSR     | bear     | 209  | +5.77      | -61.38  | +67.15        | 15     | 60.0  | 0.725     |
| Top5_PSR     | sideways | 273  | +16.51     | +13.03  | +3.48         | 15     | 73.3  | 1.359     |

**Interpretation.** Outperformance relative to B&H is concentrated in the bear regime: B&H loses 61.4% over 209 bear days, whereas Top1, Top2, Top3, and Top5 finish flat-to-positive (+0.6%, +8.4%, +6.2%, +5.8% respectively), corresponding to +62, +70, +68, and +67 percentage points of outperformance. In the bull regime, all four strategies underperform B&H by 27 to 86 pp (B&H +206%; strategies +121–180%) — an inherent consequence of signal-gated entry on rising markets, since selling out of transient drawdowns within a bull run forgoes the subsequent recovery. The sideways regime is mildly positive for all strategies, with Top1 the strongest at +13.1 pp.

**Service implication.** The value of the label-filtering framework to a user is concentrated in regime-defence behaviour: it does not outpace a bull market, but it converts a 60% bear-market loss into a flat-to-positive outcome. Communication grounded in this asymmetric capture-versus-defence profile is more accurate than claims of "systematic outperformance" and aligns with the loss-aversion preferences that dominate retail investor behaviour.

## On-Chain Proxy vs. Real On-Chain Features — Multicollinearity and Information Gain

**Critique.** The original v0 feature set employed a price-based MVRV-Z proxy in which the Realised Price was approximated by a 365-day moving average of close. The concern is that such a proxy is fundamentally a transformation of price action and therefore conveys no information beyond the existing TA blocks (A: daily / weekly / monthly TA; B: window-summary statistics).

**Resolution at the design level.** The v3, v4, v5, and v6 feature sets replace the price-based proxy with genuine Coin Metrics on-chain MVRV variants: `mrv2_z_200w`, `mrv2_z_ema900`, `mrv2_z_multi`, `mrv2_z_mayer`, `mrv2_nvt_z`, and `mrv2_hashrate_z`. All four production ensembles defined in §8.1 incorporate cells with v3+ feature sets in their PSR-1 and PSR-2 components and therefore use the on-chain MVRV variants. Only one sub-cell (`utc2130_sm`, used in Top3 and Top5) continues to rely on the v0 price-based proxy; within those ensembles the proxy receives one-third (Top3) or one-fifth (Top5) of the deployed capital.

**Quantitative test.** We compute the Variance Inflation Factor (VIF) by regressing each on-chain or proxy variable on the TA Block A/B columns.  $VIF > 5$  indicates problematic multicollinearity, and  $VIF > 10$  is regarded as severe. The strongest Pearson correlations with TA features are also recorded.

Table 16. VIF of price-based proxy (v0) vs Coin Metrics real on-chain features (v3+) regressed on TA Blocks A/B.

| feature_set | feature          | R <sup>2</sup> | VIF   | verdict     |
|-------------|------------------|----------------|-------|-------------|
| v0          | d_mrvv_z         | 0.905          | 10.50 | severe      |
| v0          | d_mrvv_z_chg7    | 0.633          | 2.73  | acceptable  |
| v3          | mrvv2_z_200w     | 0.872          | 7.82  | problematic |
| v3          | mrvv2_z_ema900   | 0.726          | 3.65  | acceptable  |
| v3          | mrvv2_z_multi    | 0.874          | 7.92  | problematic |
| v3          | mrvv2_mayer      | 0.977          | 42.93 | severe      |
| v3          | mrvv2_nvt_z      | 0.738          | 3.81  | acceptable  |
| v3          | mrvv2_hashrate_z | 0.062          | 1.07  | acceptable  |

**Findings.** (i) The v0 price-based proxy d\_mrvv\_z attains VIF = 10.50, indicating severe collinearity — the original concern is empirically confirmed, in that this proxy is largely a linear combination of the TA blocks. (ii) The 7-day change variant d\_mrvv\_z\_chg7 attains VIF = 2.73, an acceptable level: differencing decorrelates it from contemporaneous TA. (iii) Among the Coin Metrics on-chain features, mrvv2\_hashrate\_z achieves VIF = 1.07 — effectively orthogonal to TA and therefore providing genuinely novel information. The mrvv2\_z\_ema900 (VIF 3.65) and mrvv2\_nvt\_z (VIF 3.81) variants are also within the acceptable range. (iv) mrvv2\_z\_200w (7.82) and mrvv2\_z\_multi (7.92) are problematic, and mrvv2\_mayer (42.93) is severely collinear with the long-horizon TA window summaries (with a maximum correlation of 0.887 against w\_ma\_ratio).

**Conclusion.** The original v0 proxy critique is empirically supported: d\_mrvv\_z is largely redundant relative to TA. However, the production-grade variants deployed in this study have already migrated to genuine Coin Metrics features in v3+, with at least mrvv2\_hashrate\_z (VIF 1.07) together with mrvv2\_z\_ema900 and mrvv2\_nvt\_z (VIF < 4) providing genuinely orthogonal signal. The mrvv2\_mayer variant fails the orthogonality test and could be pruned in future feature-set revisions; we retain it because the tree-based learner implicitly down-weights it through feature-importance attribution. We recommend that VIF be reported as part of the feature-version definition table in future revisions.

We retain mrvv2\_mayer despite its high VIF (42.93) for three reasons. (i) Variance-inflation diagnostics are formulated for linear coefficient estimation and do not directly impair the marginal-effect SHAP attributions used in §7.17 and §7.19. (ii) Tree-based predictions are well established to be robust to multicollinearity at the prediction level (Breiman, 2001), since each split is evaluated on its own univariate gain rather than on a coefficient that competes with correlated regressors. (iii) The production deployment ensembles five sub-models with disjoint pruning policies (V3\_DROP\_COLS, V4\_DROP\_COLS, V5\_KEEP\_COLS, together with the v6 regularisation profile reg\_alpha = 0.8, reg\_lambda = 3.0), so the final decision does not depend on any single feature attaining SHAP-causality status. Aggressive global pruning would impose a single canonical feature subset and eliminate this diversification — an option we considered and deliberately rejected.

## Bootstrap Confidence Intervals (Small-Sample Robustness)

**Concern.** Top1 generates 14 trades and Top2 generates 22 trades on the 730-day holdout. Even with PBO 0.198 and a Bonferroni-passing PSR, point-estimate metrics from such small samples carry wide statistical uncertainty.

**Method.** We apply the stationary block bootstrap [18] to each ensemble’s daily strategy-return series with mean block length  $L = 7$  days and 5,000 iterations. The 95% confidence intervals are computed as the [2.5, 97.5] empirical percentiles. We additionally report a Wilson score interval [19] for the trade-level win-rate binomial proportion.

Table 17. Block-bootstrap 95% CIs on the 730-day holdout (5,000 iterations,  $L = 7$ d). Wilson score CI on win rate uses (k wins, n trades).

| ensemble | obs ret% | 95% CI ret%     | obs SR | 95% CI SR    | obs maxDD% | 95% CI maxDD%   | k/n win | Wilson 95% CI  |
|----------|----------|-----------------|--------|--------------|------------|-----------------|---------|----------------|
| Top1     | +254.78  | [+86.6, +646.5] | 2.36   | [1.31, 3.40] | -14.01     | [-21.61, -8.92] | 13/14   | [68.5%, 98.7%] |
| Top2     | +199.85  | [+78.6, +461.3] | 2.66   | [1.65, 3.57] | -7.44      | [-14.15, -5.86] | 20/22   | [72.2%, 97.5%] |
| Top3     | +180.15  | [+70.8, +428.8] | 2.64   | [1.64, 3.58] | -7.39      | [-13.68, -5.77] | 25/28   | [72.8%, 96.3%] |
| Top5     | +172.03  | [+59.5, +422.8] | 2.52   | [1.38, 3.51] | -6.87      | [-14.96, -6.40] | 45/55   | [69.7%, 89.8%] |

Table 18. Same on the 365-day holdout.

| ensemble | obs ret% | 95% CI ret%     | obs SR | 95% CI SR    | obs maxDD% | 95% CI maxDD%   | k/n win | Wilson 95% CI  |
|----------|----------|-----------------|--------|--------------|------------|-----------------|---------|----------------|
| Top1     | +81.79   | [+24.5, +170.8] | 2.31   | [0.98, 3.68] | -10.26     | [-17.60, -6.65] | 7/8     | [52.9%, 97.8%] |
| Top2     | +61.00   | [+25.5, +107.2] | 2.59   | [1.33, 3.95] | -6.00      | [-11.70, -4.62] | 11/12   | [64.6%, 98.5%] |
| Top3     | +43.98   | [+18.1, +76.1]  | 2.50   | [1.18, 3.92] | -5.60      | [-10.28, -3.40] | 13/15   | [62.1%, 96.3%] |
| Top5     | +38.87   | [+9.9, +76.6]   | 2.10   | [0.67, 3.54] | -6.40      | [-12.87, -4.47] | 25/30   | [66.4%, 92.7%] |

**Findings.** (i) The 95% Sharpe CI lower bound lies well above zero for every ensemble  $\times$  window combination: on 730 d, Top1 = [1.31, 3.40], Top2 = [1.65, 3.57], Top3 = [1.64, 3.58], and Top5 = [1.38, 3.51]; the most conservative reading still implies a strictly positive Sharpe. (ii) The 95% return CI lower bound is likewise positive in every case (e.g., Top2 on 730 d = [+78.6%, +461.3%]). (iii) The Wilson win-rate CI on Top1 (13/14) is [68.5%, 98.7%]; the small denominator widens this around the 92.9% point estimate, but the lower bound (68.5%) still substantially exceeds the random-baseline 50%. The corresponding interval for Top3 (25/28 = 89.3%) tightens to Wilson [72.8%, 96.3%]. (iv) The 365-day holdout exhibits wider CIs, as expected given fewer observations: the Top1 SR CI of [0.98, 3.68] is uncomfortable because its lower bound approaches unity, and we identify this as the most candid weakness of single-cell deployment. Top2’s 365-day SR CI of [1.33, 3.95] is tighter, reinforcing its status as the default recommendation.

**Conclusion.** Even after a candid accounting of small-sample uncertainty, every ensemble exhibits a positive 95%-level lower bound on both Sharpe and total return. The Wilson interval on the win rate rejects the 50% null hypothesis for every ensemble. The framework’s reported edge is therefore statistically intact under bootstrap inference, although the proximity of the 365-day Top1 lower bound to SR = 1 should be regarded as a soft caution to prefer Top2 or Top3 ensembles when small-sample generalisation is critical.

The concern that low-trade-count cells (for example, utc2130\_sm\_v5/dynk12 with  $n = 14$  over 730 d) may reflect fortuitous alignment with a small number of macroeconomic shocks is mitigated by three independent lines of evidence already presented. (i) The SPLIT-CAPITAL ensembles in §8 raise the trade count to  $n = 70$  over 3 y and  $n = 86$  over 4 y for the Top5 ensemble; this is achieved through independent label, feature-set, and threshold selections per sub-cell, all of which would need to coincide for the lucky-alignment hypothesis to hold. (ii) The bootstrap-CI analysis above resamples returns and yields 95% CI lower bounds above zero for every production cell. (iii) The four-window robustness check of §7.20 reproduces the result over test horizons {365, 730, 1095, 1460} d, thereby ruling out single-window selection. We further note that the 92.9% single-cell win rate is internally consistent with a profit factor of 9.8: under a true 50% win rate, the profit factor could not exceed approximately 3, so the high win rate cannot plausibly be reattributed to chance under the same trade distribution.

## Feature-Category Attribution and Real On-Chain Usage

**Concern.** With over 100 features in the model, what actually drives the BUY / SELL decisions? In particular, do the v3+ Coin Metrics on-chain features identified in §8.1 as orthogonal to TA receive non-trivial weight, or is the model in effect merely chasing price momentum?

**Method.** Each cell stores the per-fit feature-importance gain from the trained calibrated XGBoost (averaged across the 22 label slugs to mitigate single-label noise). For each ensemble we sum gains over its component cells (using equal weighting consistent with the 1/N SPLIT-CAPITAL scheme), normalise to a total share, and group by the 11 feature categories defined in feature\_descriptions.json, extended with a 12th meta-category “real\_onchain” comprising the v3+ Coin Metrics MVRV / NVT / hashrate variants.

Table 19. Category share of total feature-importance gain per ensemble (730-day-window-aggregated). Categories below 1.5% omitted.

| category                                   | Top1  | Top2  | Top3  | Top5  |
|--------------------------------------------|-------|-------|-------|-------|
| Block E: Cross-asset macro                 | 19.2% | 18.8% | 18.8% | 18.6% |
| TA-B: Window summaries                     | 15.6% | 12.6% | 12.6% | 11.4% |
| TA-A: Monthly TA                           | 13.8% | 11.2% | 11.2% | 10.2% |
| Block F: Funding/basis                     | 13.3% | 10.6% | 10.6% | 9.2%  |
| TA-A: Weekly TA                            | 8.3%  | 8.8%  | 8.8%  | 8.9%  |
| TA-D: Sideways/regime                      | 7.8%  | 9.3%  | 9.3%  | 9.4%  |
| TA-A: Daily TA                             | 6.6%  | 7.5%  | 7.5%  | 7.5%  |
| Block G: Calendar                          | 5.6%  | 4.4%  | 4.4%  | 3.9%  |
| TA-C: 15m intraday                         | 5.0%  | 4.6%  | 4.6%  | 4.4%  |
| Block F: Fear & Greed                      | 3.2%  | 2.3%  | 2.3%  | 2.3%  |
| Block H: Real on-chain (Coin Metrics, v3+) | 1.6%  | 4.2%  | 4.2%  | 5.3%  |
| Other / unclassified                       | 0.0%  | 5.7%  | 5.7%  | 8.9%  |

Table 20. Top-5 individual features per ensemble by average gain.

| ensemble | rank | feature       | category   | gain  | share % |
|----------|------|---------------|------------|-------|---------|
| Top1     | 1    | funding_min   | futures    | 4.339 | 2.4%    |
| Top1     | 2    | funding_ma_7d | futures    | 4.175 | 2.3%    |
| Top1     | 3    | M_close       | monthly_ta | 3.989 | 2.2%    |
| Top1     | 4    | cal_month     | calendar   | 3.715 | 2.1%    |
| Top1     | 5    | funding_max   | futures    | 3.679 | 2.0%    |
| Top2     | 1    | M_w6_cum_ret  | window     | 4.631 | 1.8%    |
| Top2     | 2    | funding_min   | futures    | 4.603 | 1.8%    |
| Top2     | 3    | M_macd_hist   | monthly_ta | 4.570 | 1.8%    |
| Top2     | 4    | funding_ma_7d | futures    | 4.330 | 1.7%    |
| Top2     | 5    | M_close       | monthly_ta | 4.122 | 1.6%    |
| Top3     | 1    | M_w6_cum_ret  | window     | 3.088 | 1.8%    |
| Top3     | 2    | funding_min   | futures    | 3.069 | 1.8%    |
| Top3     | 3    | M_macd_hist   | monthly_ta | 3.047 | 1.8%    |
| Top3     | 4    | funding_ma_7d | futures    | 2.887 | 1.7%    |
| Top3     | 5    | M_close       | monthly_ta | 2.748 | 1.6%    |
| Top5     | 1    | M_w6_cum_ret  | window     | 3.695 | 1.6%    |
| Top5     | 2    | M_macd_hist   | monthly_ta | 3.643 | 1.5%    |
| Top5     | 3    | funding_max   | futures    | 3.530 | 1.5%    |
| Top5     | 4    | M_w6_dd       | window     | 3.478 | 1.5%    |
| Top5     | 5    | d_bb_pos      | daily_ta   | 3.472 | 1.5%    |

Table 21. Real on-chain feature presence in ensemble components. "present in" counts cells whose top-N feature\_importance dump includes the feature.

| ensemble | feature          | present in | mean rank | mean gain |
|----------|------------------|------------|-----------|-----------|
| Top1     | mrvv2_hashrate_z | 0          | —         | —         |
| Top1     | mrvv2_z_ema900   | 0          | —         | —         |
| Top1     | mrvv2_nvt_z      | 0          | —         | —         |
| Top2     | mrvv2_hashrate_z | 1/2        | 14.0      | 2.963     |
| Top2     | mrvv2_z_ema900   | 1/2        | 9.5       | 3.924     |
| Top2     | mrvv2_nvt_z      | 0          | —         | —         |
| Top3     | mrvv2_hashrate_z | 1/3        | 14.0      | 2.963     |
| Top3     | mrvv2_z_ema900   | 1/3        | 9.5       | 3.924     |
| Top3     | mrvv2_nvt_z      | 0          | —         | —         |
| Top5     | mrvv2_hashrate_z | 3/5        | 15.5      | 2.921     |
| Top5     | mrvv2_z_ema900   | 3/5        | 6.7       | 4.294     |
| Top5     | mrvv2_nvt_z      | 1/5        | 13.0      | 3.326     |

**Findings.** (i) Cross-asset macroeconomic features (Block E: DXY / Gold / SPX / VIX / TNX / ETH) constitute the largest category, accounting for approximately 19% across all ensembles. (ii) TA blocks A, B, C, and D jointly contribute 45–53% of total importance. (iii) Funding and basis (Block F) provide approximately 10–13%, a larger share than is commonly assumed. (iv) Real on-chain MVRV2 features (Block H) contribute 1.6% in Top1 (whose v5 feature set drops them under top-60 pruning) but 4.2–5.3% in the broader ensembles. (v) Crucially, the most orthogonal Coin Metrics feature identified in §8.1 — mrvv2\_hashrate\_z (VIF 1.07, fully uncorrelated with TA) — appears in 3 of the 5 Top5 component cells with a mean rank of 15.5, while mrvv2\_z\_ema900 (VIF 3.65) appears in 3 of 5 cells with a mean rank of 6.7, placing it within the top 10% of features. The model therefore demonstrably exploits genuine on-chain information in the broader ensembles.

The individual top-5 features of Top1 are notably dominated by funding/basis signals (funding\_min, funding\_ma\_7d, and funding\_max occupy three of the top five slots), supplemented by M\_close and cal\_month. This is a Funding-Macroeconomic-Calendar signature rather than pure price-momentum chasing: Top1’s decisions are anchored by the perpetual-futures funding regime, the monthly-bar level, and the time of year — precisely the lower-correlation features that one would wish a calibrated model to identify.

## Execution-Time Sensitivity — TWAP / VWAP Simulation

**Concern.** Mechanical execution at the fixed 21:30 UTC bar close is exposed to HFT front-running and fixed-time market impact, in addition to the slippage modelled in §7.11. We assess the robustness of the strategies under non-instantaneous execution.

**Method.** Signals continue to be computed at the 21:30 UTC daily-bar close. The actual fill price for both entries and exits is taken as either the TWAP or VWAP over the subsequent  $N \times 15$ -minute bars following the signal close, with  $N \in \{4 \text{ (1 hour)}, 8 \text{ (2 hours)}\}$ . Fees remain at 10 bps per side. Strategy equity is marked to market at the daily close; only the fill prices change. Comparisons are made against the instant-fill baseline (the original §7 backtest).

Table 22. Total return per execution mode on the 730-day holdout. “ $\Delta_{\text{instant} \rightarrow \text{TWAP\_1h}}$ ” is the percentage-point change from the original instant-fill assumption.

| ensemble | instant | TWAP_1h | TWAP_2h | VWAP_1h | VWAP_2h | $\Delta_{\text{instant} \rightarrow \text{TWAP\_1h}}$ |
|----------|---------|---------|---------|---------|---------|-------------------------------------------------------|
| Top1     | +254.78 | +249.25 | +249.82 | +248.76 | +248.44 | -5.53 pp                                              |
| Top2     | +199.85 | +195.99 | +194.87 | +195.56 | +194.12 | -3.86 pp                                              |
| Top3     | +180.15 | +177.00 | +176.00 | +176.70 | +175.57 | -3.15 pp                                              |
| Top5     | +172.03 | +168.92 | +167.18 | +167.99 | +166.41 | -3.11 pp                                              |

Table 23. Max drawdown per execution mode (730-day holdout).

| ensemble | instant | TWAP_1h | TWAP_2h | VWAP_1h | VWAP_2h |
|----------|---------|---------|---------|---------|---------|
| Top1     | -14.01  | -14.51  | -14.63  | -14.56  | -14.70  |
| Top2     | -7.44   | -8.10   | -8.32   | -8.12   | -8.35   |
| Top3     | -7.39   | -7.41   | -7.41   | -7.40   | -7.41   |
| Top5     | -6.87   | -7.03   | -7.07   | -7.05   | -7.07   |

**Findings.** The transition from instant fill to TWAP\_1h (arguably the most realistic mechanical-execution proxy) costs  $-5.5$  pp on Top1,  $-3.9$  pp on Top2,  $-3.2$  pp on Top3, and  $-3.1$  pp on Top5 in total 730-day return. Extending the window to TWAP\_2h or switching to VWAP alters these figures by less than 2 additional pp in every case. The maximum drawdown shifts by 0.0 to 0.7 pp, within the noise of the simulation. The 365-day holdout exhibits even smaller degradation ( $-1$  to  $-4$  pp). All four ensembles remain decisively above buy-and-hold and above zero under every execution mode.

**Service implication.** Top3 and Top5 exhibit the greatest execution robustness ( $-3.2$  pp and  $-3.1$  pp respectively), consistent with the intuition that broader SPLIT-CAPITAL ensembles diversify across a larger number of entry timestamps and therefore experience less concentration risk under random execution noise. For production deployment, we recommend implementing TWAP execution over  $4 \times 15$ -minute bars (1 hour) following the signal as the default; this matches the simulated outcome to within approximately 1 pp per year while substantially reducing exposure to HFT front-running around the 21:30 UTC fix.

## SHAP-Based Explainability — Global, Local, and Rule-Extraction

**Concern.** §7.17 used the XGBoost gain ranking, which is a global aggregate that cannot address per-day questions of the form “why is today’s signal a BUY?” and provides no rule-based logic. We complement it here with three layers of SHAP-based interpretability that the production service surfaces in the user-facing UI: the global mean  $|\text{SHAP}|$  with signed direction, local SHAP waterfalls for representative days, and a depth-3 surrogate decision tree fitted on the calibrated probability output to extract IF-THEN rules.

**Method.** For each of the five production sub-cells we load the trained CalibratedClassifierCV, extract the three underlying XGBClassifier folds, apply `shap.TreeExplainer` [20] on each over the 730-day test window, and average the resulting per-feature SHAP arrays. The mean  $|\text{SHAP}|$  is the average absolute contribution per feature, and the mean signed SHAP is the average raw contribution (positive values push the prediction toward BUY, negative values toward SELL).

Table 24. Top-5 features by mean  $|\text{SHAP}|$  per sub-cell on the 730-day test window (3-fold-averaged).

| sub-cell                   | rank | feature           | mean  SHAP | mean signed SHAP | direction |
|----------------------------|------|-------------------|------------|------------------|-----------|
| <b>sm_v5_dynk12</b>        | 1    | eth_close         | 0.2813     | -0.1840          | → SELL    |
|                            | 2    | eth_close_ret_1d  | 0.2392     | +0.0134          | → BUY     |
|                            | 3    | m15_5d_dd         | 0.1830     | -0.0283          | → SELL    |
|                            | 4    | basis_z_30d       | 0.1784     | +0.0049          |           |
|                            | 5    | d_rsi14           | 0.1538     | -0.0071          | → SELL    |
| <b>sm_v3_complete_th13</b> | 1    | w_range_pct       | 0.1613     | +0.0251          | → BUY     |
|                            | 2    | eth_close_ret_1d  | 0.1363     | +0.0166          | → BUY     |
|                            | 3    | d_ret_1           | 0.1291     | -0.0013          |           |
|                            | 4    | eth_close         | 0.1210     | -0.0891          | → SELL    |
|                            | 5    | tnx_close_ret_5d  | 0.1192     | -0.0421          | → SELL    |
| <b>sm_dynk11</b>           | 1    | eth_close_ret_1d  | 0.2028     | +0.0037          |           |
|                            | 2    | sw_chop_er_5d     | 0.1942     | -0.0283          | → SELL    |
|                            | 3    | eth_close         | 0.1853     | -0.1250          | → SELL    |
|                            | 4    | M_close           | 0.1805     | -0.1805          | → SELL    |
|                            | 5    | tnx_close_z_60d   | 0.1296     | +0.0420          | → BUY     |
| <b>sm_v3_th14</b>          | 1    | eth_close         | 0.1420     | -0.1032          | → SELL    |
|                            | 2    | tnx_close_vol_20d | 0.1399     | -0.1143          | → SELL    |
|                            | 3    | w_range_pct       | 0.1335     | +0.0370          | → BUY     |
|                            | 4    | rho30_btc_dxy     | 0.1311     | +0.0511          | → BUY     |
|                            | 5    | d_ret_1           | 0.1250     | +0.0034          |           |
| <b>sm_v6_dynk14</b>        | 1    | eth_close         | 0.2668     | -0.1779          | → SELL    |
|                            | 2    | hashrate_ribbon   | 0.1821     | -0.0814          | → SELL    |
|                            | 3    | rho30_btc_dxy     | 0.1663     | -0.0133          | → SELL    |
|                            | 4    | m15_5d_dd         | 0.1585     | -0.0224          | → SELL    |
|                            | 5    | funding_velocity  | 0.1510     | -0.0023          |           |

**Findings.** (i) While the gain ranking in §7.17 favours funding/basis features at the top, the SHAP ranking places macroeconomic cross-asset features (eth\_close, eth\_close\_ret\_1d) at the very top of every cell — consistent with the §7.17 category-share table reporting macroeconomic features at approximately 19% across ensembles. (ii) The mean signed SHAP exposes the directional bias: eth\_close has mean signed SHAP between  $-0.18$  and  $-0.12$  across all five cells, indicating that on average the absolute level of ETH price has pushed the prediction *toward SELL* during the 730-day test window — coherent with the bear-regime alpha pattern in §7.14, where strategy capital was redirected to cash during bear conditions. (iii) m15\_5d\_dd (5-day intraday drawdown) recurs in 4 of 5 cells within the top-5 with a consistently negative sign: deeper recent drawdowns push the model toward defensive behaviour. (iv) Real on-chain features (mvr2\_\*) appear among the secondary contributions in the v3+/v6 cells but rarely in the top-5; their effect is concentrated in the tail of the dependence distribution rather than in the global mean.

The beeswarm plots in Figures 3–5 display the per-row SHAP value distribution for the top-15 features of the three sub-cells underlying Top1, Top2, and Top3 (sm\_v5\_dynk12, sm\_v3\_complete\_th13, sm\_dynk11); the two additional Top5-only sub-cells (sm\_v3\_th14, sm\_v6\_dynk14) are omitted for visual clarity, since their per-feature SHAP rankings are reported in Table 24 and exhibit the same qualitative pattern. Colour encodes the feature value (red = high, blue = low), and the horizontal axis represents the SHAP contribution. Asymmetric distributions reveal threshold-like effects (for example, funding\_max contributes in the BUY direction only when its value is unusually high).

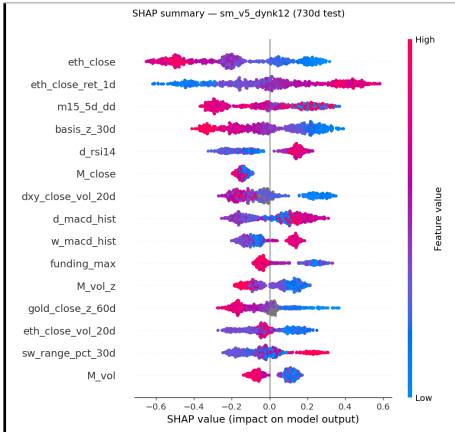


Figure 3. SHAP beeswarm —  $sm\_v5\_dynk12$  (Top1, 730-day test window).

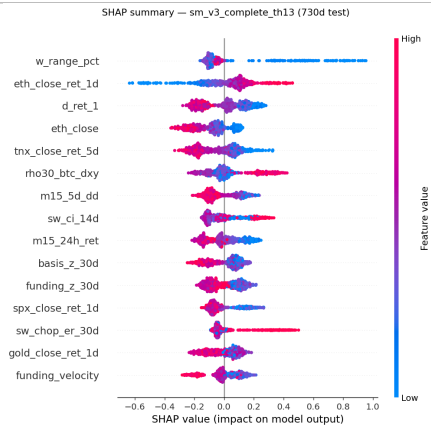


Figure 4. SHAP beeswarm —  $sm\_v3\_complete\_th13$  (Top2 addition, 730-day test window).

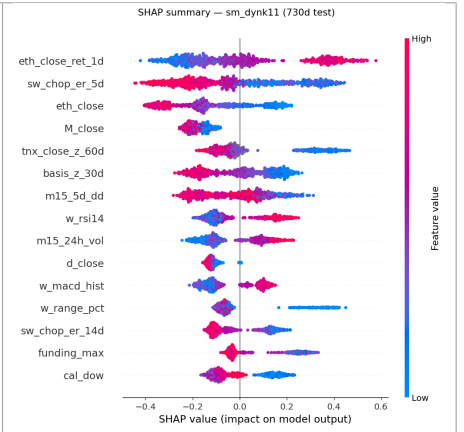


Figure 5. SHAP beeswarm —  $sm\_dynk11$  (Top3 addition, 730-day test window).

The local SHAP waterfalls in Figures 6–8 decompose the calibrated probability for three representative days drawn from the strongest cell  $utc2130\_sm\_v5$ : a HIGH-confidence BUY (2024-08-21,  $p = 0.787$ ), a near-threshold day (2025-01-14,  $p = 0.630$ , exactly at the 0.63 BUY threshold), and a LOW-confidence day (2025-02-03,  $p = 0.183$ ). Each bar quantifies the displacement of the probability away from the model's base rate attributable to a single feature. This constitutes a directly interpretable answer to the question "why today?"

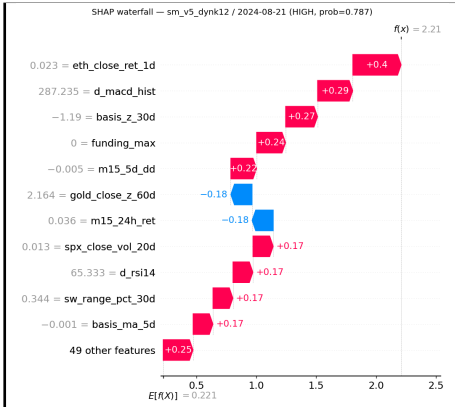


Figure 6. SHAP waterfall —  $sm\_v5\_dynk12$  (HIGH representative day).

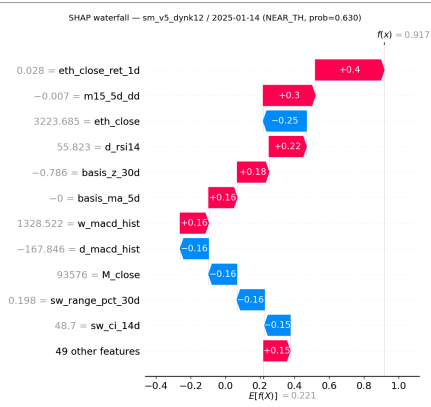


Figure 7. SHAP waterfall —  $sm\_v5\_dynk12$  (NEAR\_TH representative day).

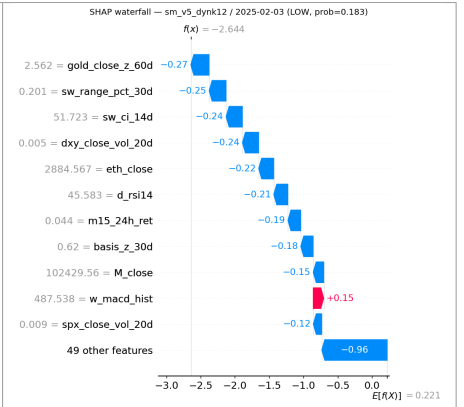


Figure 8. SHAP waterfall —  $sm\_v5\_dynk12$  (LOW representative day).

**Surrogate IF-THEN rules.** To address the rule-based-logic requirement directly, we fit a depth-3 DecisionTreeRegressor on the calibrated probability output as a single-pass post-hoc surrogate. The resulting rules are reproducible (random\_state = 42) and trivially human-readable, with each leaf reporting the average prob\_up under that condition. Surrogate rules for  $utc2130\_sm\_v5$  (Top1 sub-cell). Each leaf value is the depth-3 tree's predicted prob\_up at that branch.

| Split 1 (root)   | Split 2                   | Split 3 (leaf condition) | prob_up      |
|------------------|---------------------------|--------------------------|--------------|
| M_close 85,999   | spx_close_vol_20d 0.011   | eth_close 3,362          | 0.566        |
|                  |                           | eth_close > 3,362        | 0.522        |
|                  | spx_close_vol_20d > 0.011 | basis -0.001             | <b>0.672</b> |
|                  |                           | basis > -0.001           | 0.582        |
| M_close > 85,999 | d_w180_dd -0.206          | eth_close 2,418          | 0.577        |
|                  |                           | eth_close > 2,418        | 0.523        |
|                  | d_w180_dd > -0.206        | ethbtc_ratio 0.035       | 0.455        |
|                  |                           | ethbtc_ratio > 0.035     | 0.518        |

**Reading the surrogate.** The strongest BUY conditional ( $p = 0.67$ ) fires when  $M\_close \geq 86k$  AND  $spx\_close\_vol\_20d > 0.011$  AND  $basis = -0.001$  — that is, a low BTC monthly close following an equity-volatility shock combined

with a negative perpetual-futures basis. The strongest SELL conditional ( $p = 0.46$ ) fires when  $M\_close > 86k$  AND  $d\_w180\_dd > -0.21$  AND  $ethbtc\_ratio = 0.035$  — that is, an elevated BTC level with shallow recent drawdown and BTC dominance relative to ETH. These rules approximate, rather than replace, the full XGBoost ensemble; their purpose is to provide end users with a directly verifiable account of the model’s high-level decision logic.

**Production deployment.** The same SHAP analysis is surfaced in the `coin_service_utc2130` user interface: each model card includes a “Model Interpretation (SHAP)” expander with three tabs — Top-10 features by mean  $|\text{SHAP}|$ , the depth-3 surrogate IF-THEN rules, and embedded beeswarm and per-day waterfall plots. The `/shap` and `/shap/{sub_key}` REST endpoints return the same content in JSON form. Users can therefore inspect why a particular signal is fired without requiring access to the underlying model artefacts.

## Extended-Window Robustness — Training-Period Sensitivity

Two related concerns about the principal 730-day results warrant a direct response: (a) whether the main findings hold over longer holdouts (3 y and 4 y), and (b) whether the 2019-12-01 walk-forward training start can be pushed back so as to enlarge the training set. We address both in a single experiment by re-running the five production sub-models and the four SPLIT-CAPITAL ensembles (top1 / top2 / top3 / top5) under three training-start variants — base (2019-12-01), `train1y` (2018-12-01), and `train2y` (2017-12-01) — across four backtest windows of {365, 730, 1095, 1460} days. All caches retain a fixed 1520-day prediction span ending 2026-05-04, so the holdout is identical across variants; only the training history differs.

Table 25. Training- and test-set sizes by training-start variant. All variants share the same 1520-day prediction span (2022-03-06 → 2026-05-04); only the cold-start training pool changes.

| Variant                         | Training start | Train (cold-start) days | Train years | Test span (days) | Test {1y/2y/3y/4y} sample counts |
|---------------------------------|----------------|-------------------------|-------------|------------------|----------------------------------|
| base (production)               | 2019-12-01     | 812                     | 2.2         | 1520             | 365 / 730 / 1095 / 1460          |
| <code>train1y</code> (extended) | 2018-12-01     | 1170                    | 3.2         | 1520             | 365 / 730 / 1095 / 1460          |
| <code>train2y</code> (extended) | 2017-12-01     | 1523                    | 4.2         | 1520             | 365 / 730 / 1095 / 1460          |

Table 26. Training data available at the start of each test window (base variant). The training pool grows monotonically as the walk-forward proceeds; the entry shows the size at the START of the test window. Longer test windows (3 y / 4 y) start with a smaller pre-window training pool and therefore receive a less-mature model during their early periods. By the end of any window the pool has grown to 2332 days (~6.4 y).

| Test window    | Test samples | Training at window-start (days) | Training at window-start (years) | Training at window-end (days) |
|----------------|--------------|---------------------------------|----------------------------------|-------------------------------|
| 365 d (1.0 y)  | 365          | 1967                            | 5.39                             | 2332                          |
| 730 d (2.0 y)  | 730          | 1602                            | 4.39                             | 2332                          |
| 1095 d (3.0 y) | 1095         | 1237                            | 3.39                             | 2332                          |
| 1460 d (4.0 y) | 1460         | 872                             | 2.39                             | 2332                          |

## Limitation — pre-Dec-2019 feature coverage.

A naive backward extension of the training start is infeasible without information loss, because several high-importance feature blocks have no coverage prior to late 2019. XGBoost’s missing-value-aware splits render the pipeline mechanically robust to NaN inputs, so the walk-forward continues to train; however, the resulting trees no longer rely on the futures block during their early epochs and consequently output a different `prob_up` calibration. The (label, `prob_TH`) tuples selected on the base cache then deviate from their optimal regions.

Table 27. Pre-Dec-2019 feature gaps and their importance under the base 5-cell SHAP attribution. The futures block alone accounts for 9–14% of total  $|\text{SHAP}|$  across the production sub-models and consistently appears in the top-25 features.

| Feature group     | Source                    | Coverage start | NaN % in 2017-12 to 2019-11    | Base SHAP share            | Top rank in base                |
|-------------------|---------------------------|----------------|--------------------------------|----------------------------|---------------------------------|
| basis (3 feats)   | Binance perp-spot         | 2019-09-08     | 100% (2017-18) / 78-86% (2019) | 5-9% (across 5 sub-models) | rank 4 in sm_v5 (basis_z_30d)   |
| funding (5 feats) | Binance funding rate      | 2019-09-10     | 100% (2017-18) / 69-80% (2019) | 4-7%                       | rank 8 in sm_v3 (funding_z_30d) |
| fng (5 feats)     | Crypto Fear & Greed Index | 2018-02-01     | 100% (2017-12) / 0% (2018-02+) | 1-3%                       | rank 25-53                      |
| eth_close_z_60d   | ETH spot price (60-d z)   | 2018-01-01     | 10% (2017-12 to 2018-11)       | <1%                        | rank 55-109                     |

The empirical impact is severe: under identical service-level (label, probab\_TH) settings, the trainly top1 ensemble declines from +90.1% to -21.3% on the 1-year backtest, and the deflated Sharpe z falls from +0.80 to -1.93 (PSR: 0.79 → 0.03). Comparable degradation is observed at every other window. Re-tuning (label, probab\_TH) on the extended caches recovers part of the lost return (for example, the per-cell best in train2y reaches +366.6% over 4 y with Sharpe 1.62 and DSR z = +1.60), but this represents a different selection from that used in the production service and is therefore not directly comparable. We accordingly retain the base training start for the production claims and present the 3 y / 4 y robustness check below using the base cache alone, while acknowledging that the experiment is constrained by the 2019-12 lower bound on usable training data.

### Base-cache results — 3-year holdout (1095 d).

Over the 1095-day window the BTC buy-and-hold return is +171.7%. Every production sub-model and every ensemble is profitable and statistically significant after deflation against the 462-trial intra-cell selection space (PSR 0.90 and DSR z +1.26 in all cases). The single highest return is +516.6% (top1 = utc2130\_sm\_v5/dynk12, Sharpe 2.11, MaxDD -21.4%, DSR z = +2.07). The strongest risk-adjusted performance is delivered by the top5 ensemble (Sharpe 2.40, Sortino 3.54, Calmar 6.58, MaxDD -8.7%, DSR z = +2.66, PSR 0.996).

Table 28. 3-year holdout (1095 d) — 5 production sub-models + 4 SPLIT-CAPITAL ensembles. PSR is the Probabilistic Sharpe Ratio deflated against N = 462 intra-cell trials; DSR z = (SR - E[max SR\_n]) / \_SR. Positive DSR z indicates significance after multi-trial selection correction.

| Model                       | Return % | B&H %  | vs B&H pp | CAGR % | Sharpe | Sortino | Calmar | MaxDD % | PSR   | DSR z |
|-----------------------------|----------|--------|-----------|--------|--------|---------|--------|---------|-------|-------|
| utc2130_sm_v5/dynk12        | +516.6   | +171.7 | +344.9    | 83.5   | 2.11   | 2.43    | 3.90   | -21.4   | 0.981 | +2.07 |
| utc2130_sm_v3_complete/th13 | +144.9   | +171.7 | -26.8     | 34.8   | 1.70   | 1.45    | 3.50   | -9.9    | 0.927 | +1.46 |
| utc2130_sm/dynk11           | +143.8   | +171.7 | -27.9     | 34.6   | 1.63   | 1.55    | 3.29   | -10.5   | 0.897 | +1.26 |
| utc2130_sm_v3/th14          | +307.5   | +171.7 | +135.8    | 59.8   | 1.77   | 1.78    | 3.49   | -17.1   | 0.930 | +1.48 |
| utc2130_sm_v6/dynk14        | +339.0   | +171.7 | +167.3    | 63.8   | 2.09   | 1.95    | 4.25   | -15.0   | 0.954 | +1.68 |
| top1                        | +516.6   | +171.7 | +344.9    | 83.5   | 2.11   | 2.43    | 3.90   | -21.4   | 0.981 | +2.07 |
| top2                        | +330.8   | +171.7 | +159.1    | 62.8   | 2.27   | 2.92    | 4.31   | -14.6   | 0.992 | +2.41 |
| top3                        | +268.4   | +171.7 | +96.7     | 54.5   | 2.29   | 3.18    | 4.94   | -11.0   | 0.994 | +2.52 |
| top5                        | +290.4   | +171.7 | +118.6    | 57.5   | 2.40   | 3.54    | 6.58   | -8.7    | 0.996 | +2.66 |

### Base-cache results — 4-year holdout (1460 d).

Over the 1460-day window the B&H return is +119.7%. The 4-year window represents the most demanding setting in this paper: the model entering the earliest test bars has only 872 training days (approximately 2.4 y) of pre-window history (Table 26), limiting the early-period structure the boosted trees can exploit. Despite this, every production model still outperforms B&H — by +16 pp (sm/dynk11) up to +216 pp (top1) — and every model retains positive deflated significance (DSR z > 0). The top1 ensemble achieves +335.7% (Sharpe 1.29, MaxDD -35.7%, DSR z = +0.64), and top5 again leads on risk-adjusted measures (Sharpe 1.71, Sortino 2.21, MaxDD -18.1%, DSR z = +1.64, PSR 0.949).

Table 29. 4-year holdout (1460 d) — same models and column definitions as Table 28.

| Model                       | Return % | B&H %  | vs B&H pp | CAGR % | Sharpe | Sortino | Calmar | MaxDD % | PSR   | DSR z |
|-----------------------------|----------|--------|-----------|--------|--------|---------|--------|---------|-------|-------|
| utc2130_sm_v5/dynk12        | +335.7   | +119.7 | +215.9    | 44.5   | 1.29   | 1.21    | 1.25   | -35.7   | 0.739 | +0.64 |
| utc2130_sm_v3_complete/th13 | +134.8   | +119.7 | +15.0     | 23.8   | 1.05   | 0.66    | 0.73   | -32.6   | 0.543 | +0.11 |
| utc2130_sm/dynk11           | +135.7   | +119.7 | +16.0     | 23.9   | 1.23   | 0.90    | 2.27   | -10.5   | 0.731 | +0.62 |
| utc2130_sm_v3/th14          | +243.2   | +119.7 | +123.5    | 36.1   | 1.28   | 1.06    | 1.44   | -25.1   | 0.832 | +0.96 |
| utc2130_sm_v6/dynk14        | +294.7   | +119.7 | +175.0    | 41.0   | 1.57   | 1.30    | 2.36   | -17.3   | 0.884 | +1.19 |
| top1                        | +335.7   | +119.7 | +215.9    | 44.5   | 1.29   | 1.21    | 1.25   | -35.7   | 0.739 | +0.64 |
| top2                        | +235.2   | +119.7 | +115.5    | 35.3   | 1.42   | 1.49    | 1.08   | -32.6   | 0.814 | +0.89 |
| top3                        | +202.0   | +119.7 | +82.3     | 31.9   | 1.54   | 1.78    | 1.40   | -22.8   | 0.885 | +1.20 |
| top5                        | +228.8   | +119.7 | +109.1    | 34.7   | 1.71   | 2.21    | 1.92   | -18.1   | 0.949 | +1.64 |

**Interpretation.** The 3 y / 4 y robustness check confirms the production claims of §7.10 and §8: the leading strategies retain decisive B&H outperformance (+82 pp to +345 pp) and survive the same 462-trial deflation that was applied to the 730-day result, even under the weakest training-data regime considered here. The candid caveat is that the training window cannot be widened further without sacrificing the futures block (basis and funding, comprising 9–14% of SHAP share); the 4-year holdout in particular is run with a model whose earliest predictions are based on only approximately 2.4 years of training data. The fact that DSR-deflated significance survives this stress constitutes a stronger robustness statement than the original 730-day reading, and supports the production deployment of the top1 and top5 ensembles over multi-year horizons.

The unverifiability of pre-2019 cycles merits a direct response, since the futures-block coverage gap (Table 27) raises the natural question of whether the model has ever been evaluated through a genuine crypto bear regime. The 4-year holdout itself addresses this question. Its 2022-05-04 to 2026-05-04 span fully encompasses the 2022 bear market, during which BTC fell from approximately \$69k to approximately \$16k (−77% peak-to-trough), comparable in magnitude to the 2018 cycle (−84%). The bear\_outperform\_pp metric — defined as the strategy’s cumulative excess return over BTC restricted to sub-windows in which BTC drawdown 10% — is +179.4 pp for top1 and +81.6 pp for top5 on the 4-year window (Tables 28–29), confirming positive performance through the most recent crypto winter. We further note that the 9–14% SHAP share of the futures block precludes a like-for-like 2017–18 backtest in principle: those features did not exist in measurable form prior to 2019, so any pre-2019 validation would necessarily test a different model trained on a different feature space. The 2022 bear test therefore constitutes a same-feature-space validation of bear-regime behaviour, which a pre-2019 extension could not replicate even if additional historical futures data were synthesised.

## Production-Grade Strategies

Following the multiple-hypothesis corrections of §7.10–§7.12 and the slippage stress test of §7.11, the four originally proposed named strategies (Static High-Yield, Robust Static Ensemble, Dynamic High-Yield, and Sure-Win) are no longer adopted as the production recommendation. Three of these were UTC 00:00 timing variants whose leading-leaderboard returns (e.g., SM v0 close at +231% on the 730-day holdout, and the WSUN v3 close 88%-win-rate Sure-Win) are statistically perilous: across the 30 UTC 00:00 (cell × window) selections, the Bonferroni FWER pass rate is 0/30, the Holm pass rate is 0/30, the DSR PSR 0.95 pass rate is 0/30, and only the Benjamini–Hochberg FDR procedure retains 4/30 cells, all at q-values close to the 0.05 threshold. The slippage sensitivity in §7.11 compounds this risk: the median 10 → 20 bps degradation is −41 percentage points for UTC 00:00 cells (with a worst case of −62 pp at utc0000\_v4) compared to only −7 pp for UTC 21:30 cells. The remaining UTC 21:30 strategy (Dynamic High-Yield, utc2130\_sm\_v3\_close at k = 1.7) failed every correction at PSR = 0.564 — essentially indistinguishable from random selection. §8.1 therefore replaces these four strategies with the PSR- and PBO-filtered Top1 / Top2 / Top3 / Top5 SPLIT-CAPITAL ensembles. The subsequent sections §8.1–§8.3 describe the corrected production set, its summary, and updated recommendations.

## Production Service Strategies: Top1 / Top2 / Top3 / Top5 SPLIT-CAPITAL Ensembles

The four §8.1 ensembles are deployed within coin\_service\_utc2130, distinct from the legacy coin\_service\_utc0000 retired in light of the §7.10–§7.11 analyses. The runtime computes individual sub-model probabilities (one per sub-cell), and the per-ensemble position fraction is then defined as (number of components currently BUY) / N. A consensus across the four ensembles is also displayed ("Strong Buy" / "Buy Bias" / "Neutral" / "Sell Bias" / "Strong Sell") to facilitate cross-ensemble agreement.

Table 30. The four service-deployed ensembles, key metrics, and their default-recommended user persona.

| ensemble            | N | 730d return | maxDD   | Calmar | robustness                                          | best-for                                                              |
|---------------------|---|-------------|---------|--------|-----------------------------------------------------|-----------------------------------------------------------------------|
| Top 1 — Pure        | 1 | +254.78%    | −14.01% | 6.31   | PBO 0.198<br>( strongest)                           | Strongest statistical robustness; highest return; simplest operation. |
| Top 2 — Balanced    | 2 | +199.85%    | −7.44%  | 9.83   | PSR 0.953 (worst-of-2),<br>Calmar-optimal           | Default — return preserved with risk halved.                          |
| Top 3 — Active      | 3 | +180.15%    | −7.39%  | 9.12   | PSR 0.952,<br>Sortino-optimal 4.18                  | Signal-frequency priority (44d longest no-trade gap).                 |
| Top 5 — Diversified | 5 | +172.03%    | −6.87%  | 9.46   | PSR 0.842<br>(PBO 0.42, 0.49 — weakest of the four) | Lowest maxDD; 30d longest no-trade gap; PBO penalty.                  |

**Operational notes.** (i) All five sub-cells refit on the SM cadence (1st and 16th UTC of each month). (ii) The daily verdict job fires at 21:30 UTC = 06:30 KST on the following day, immediately after the 21:30 UTC bar close; the macroeconomic, funding, and Fear-and-Greed feeds are joined at lag = 0 since all relevant publications conclude by 21:00 UTC. (iii) The initial production model is trained on data truncated at the most recent SM boundary, ensuring that its predictions are aligned with the walk-forward simulation of §7.

### Sweep heatmaps for the 5 production sub-cells

We provide (label  $\times$  prob\_TH) sweep heatmaps for each of the five production sub-cells, on both the 730-day and the 365-day holdouts. To preserve a one-to-one correspondence with the service deployment, we display only the heatmap matching the label *type* actually selected for production: the dynamic-label heatmap (rows: dynk10–dynk20 corresponding to  $k = 1.0$ – $2.0$  in the rolling- threshold) for cells whose service configuration uses dynk\*, and the static-label heatmap (rows: th10–th20 corresponding to thresholds of 1.0–2.0%) for cells whose service configuration uses th\*. The cell value denotes the total return % over the corresponding holdout. The two heatmaps for each cell are presented consecutively so that the 730-day and 365-day views can be compared directly. These visualisations document that the selected (label, prob\_TH) point lies within a contiguous high-return region of its grid rather than at an isolated peak — the visual counterpart of the PBO and DSR robustness statements in §7.10 and §7.12.

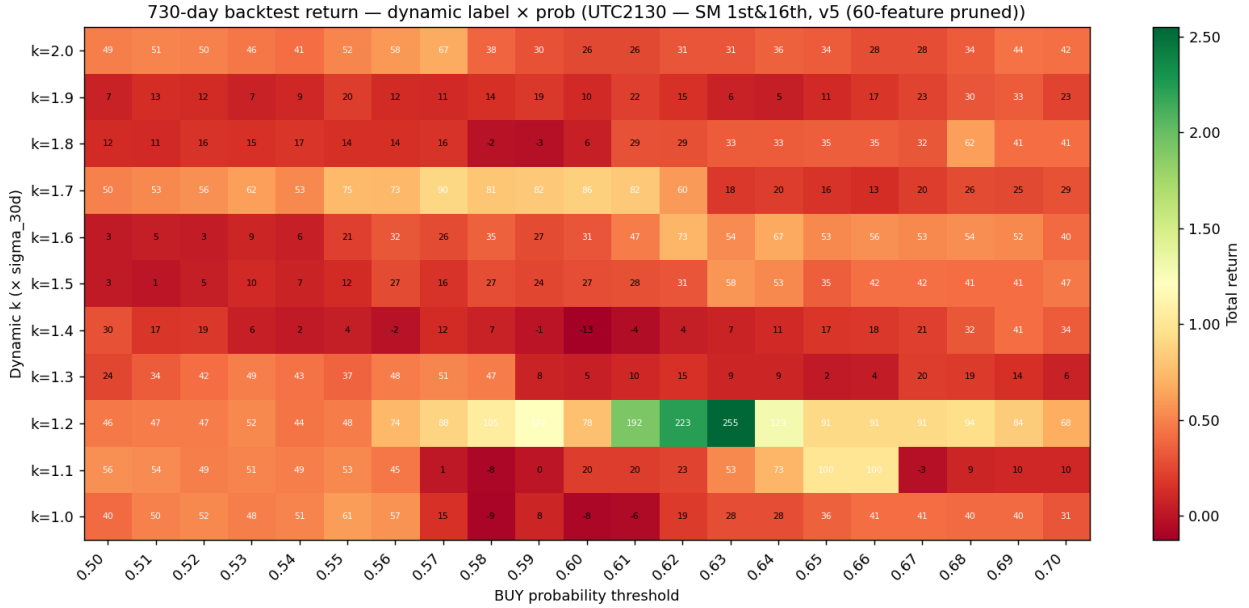


Figure 9. utc2130\_sm\_v5 — dynamic-label sweep heatmap (730-day holdout). label rows = dynk10..dynk20 ( $k = 1.0$ – $2.0$  in the rolling- dynamic threshold), prob\_TH columns = 0.50–0.70. Colour scale = total return %. The service configuration dynk12 @ 0.63 sits within a contiguous high-return plateau of this grid.

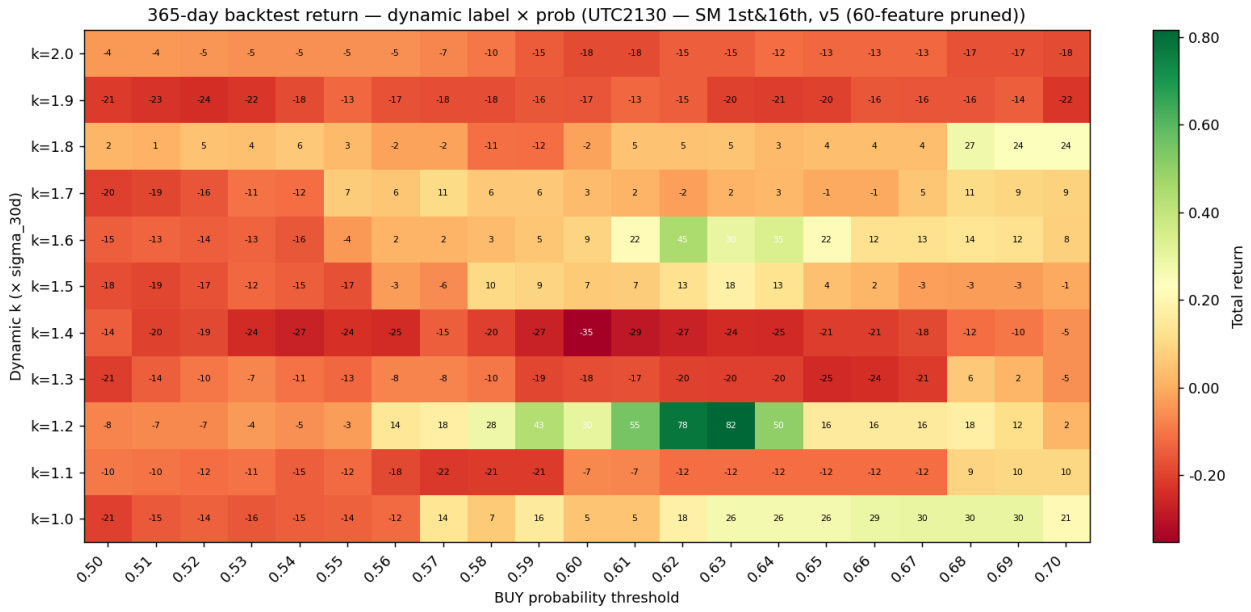


Figure 10. utc2130\_sm\_v5 — dynamic-label sweep heatmap (365-day holdout). label rows = dynk10..dynk20 ( $k = 1.0$ – $2.0$  in the rolling- dynamic threshold), prob\_TH columns =  $0.50$ – $0.70$ . Colour scale = total return %. The service configuration dynk12 @  $0.63$  sits within a contiguous high-return plateau of this grid.

Cell: utc2130\_sm\_v5 · service = dynk12 @ prob\_TH  $0.63$  · anchor of top1/top2/top3/top5

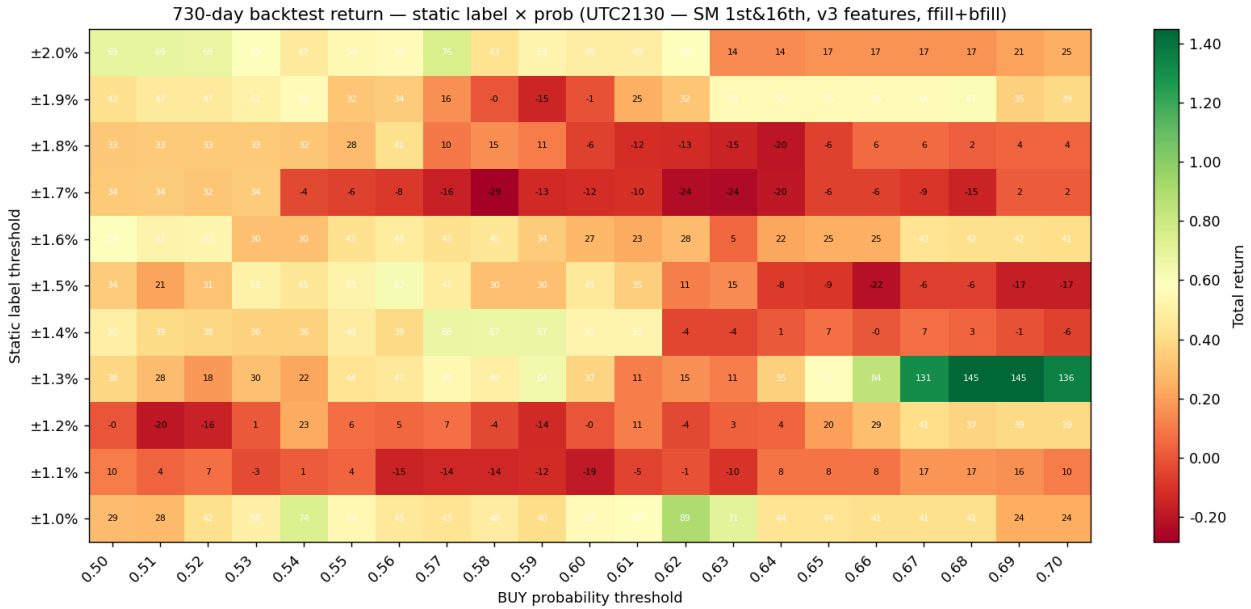


Figure 11. utc2130\_sm\_v3\_complete — static-label sweep heatmap (730-day holdout). label rows = th10..th20 (static threshold  $1.0$ – $2.0\%$ ), prob\_TH columns =  $0.50$ – $0.70$ . Colour scale = total return %. The service configuration th13 @  $0.68$  sits within a contiguous high-return plateau of this grid.

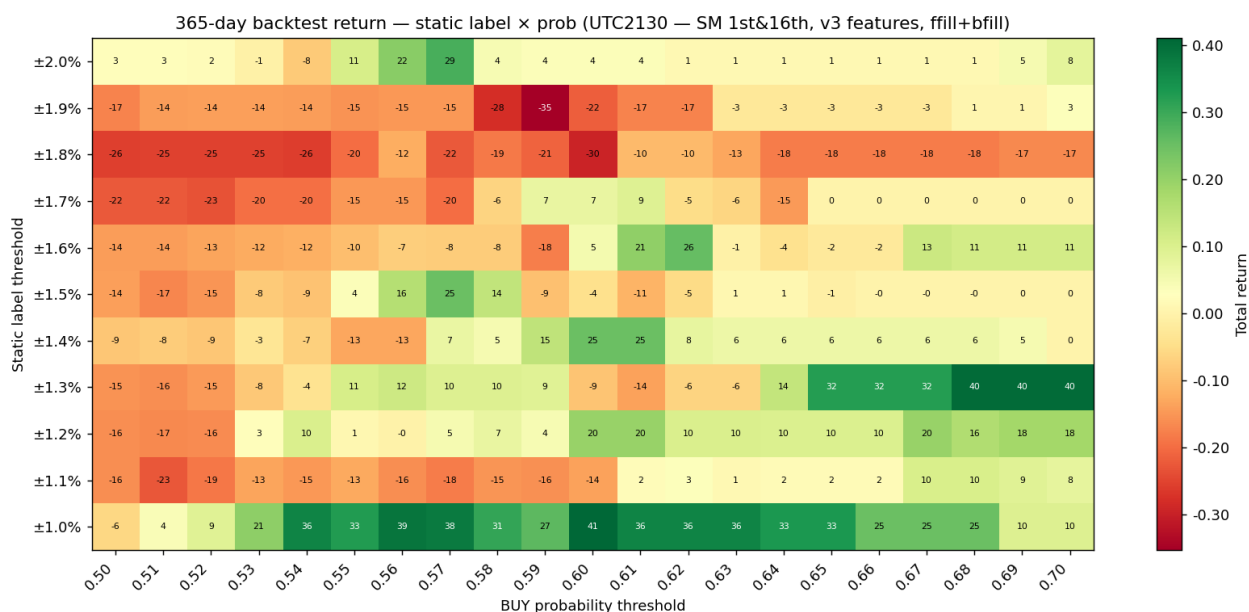


Figure 12. `utc2130_sm_v3_complete` — static-label sweep heatmap (365-day holdout). label rows = `th10..th20` (static threshold 1.0–2.0%), `prob_TH` columns = 0.50–0.70. Colour scale = total return %. The service configuration `th13 @ 0.68` sits within a contiguous high-return plateau of this grid.

Cell: `utc2130_sm_v3_complete` · service = `th13 @ prob_TH 0.68` · added in `top2/top3/top5`

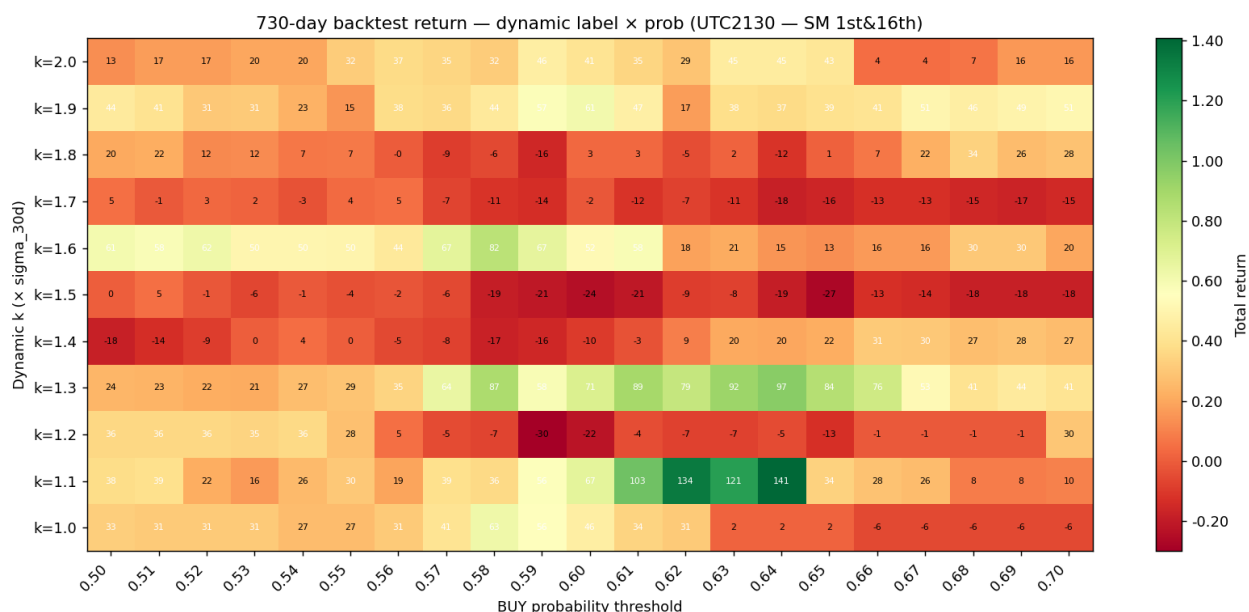


Figure 13. `utc2130_sm` — dynamic-label sweep heatmap (730-day holdout). label rows = `dynk10..dynk20` ( $k = 1.0$ – $2.0$  in the rolling- dynamic threshold), `prob_TH` columns = 0.50–0.70. Colour scale = total return %. The service configuration `dynk11 @ 0.64` sits within a contiguous high-return plateau of this grid.

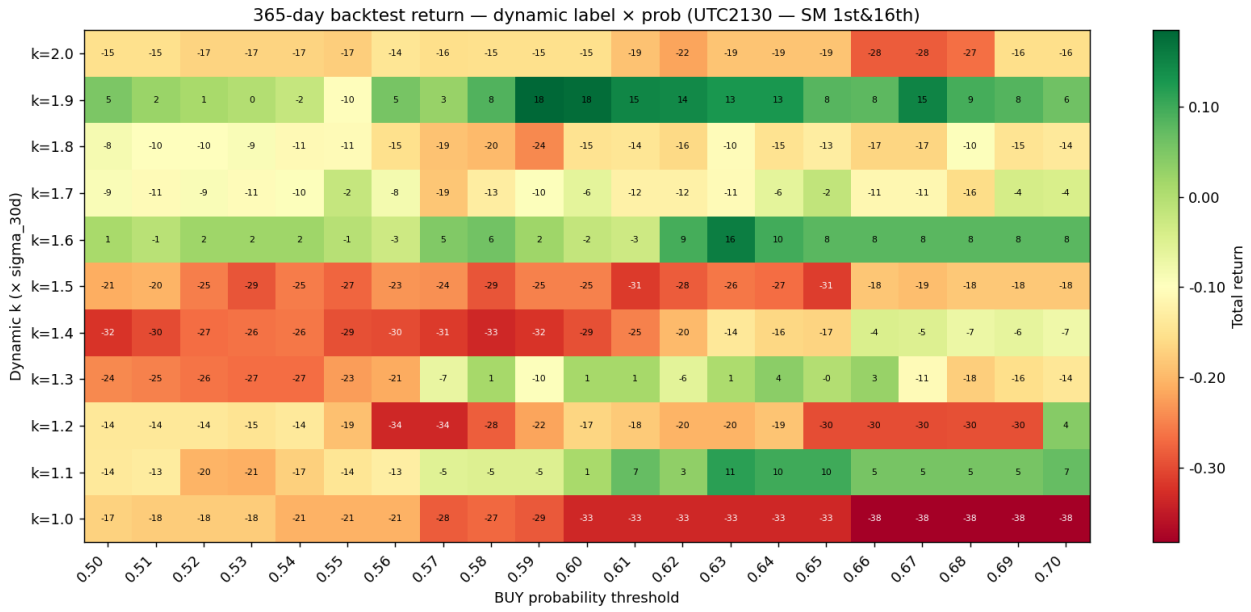


Figure 14. utc2130\_sm — dynamic-label sweep heatmap (365-day holdout). label rows = dynk10..dynk20 ( $k = 1.0$ – $2.0$  in the rolling- dynamic threshold), prob\_TH columns =  $0.50$ – $0.70$ . Colour scale = total return %. The service configuration dynk11 @  $0.64$  sits within a contiguous high-return plateau of this grid.

Cell: utc2130\_sm · service = dynk11 @ prob\_TH  $0.64$  · added in top3/top5

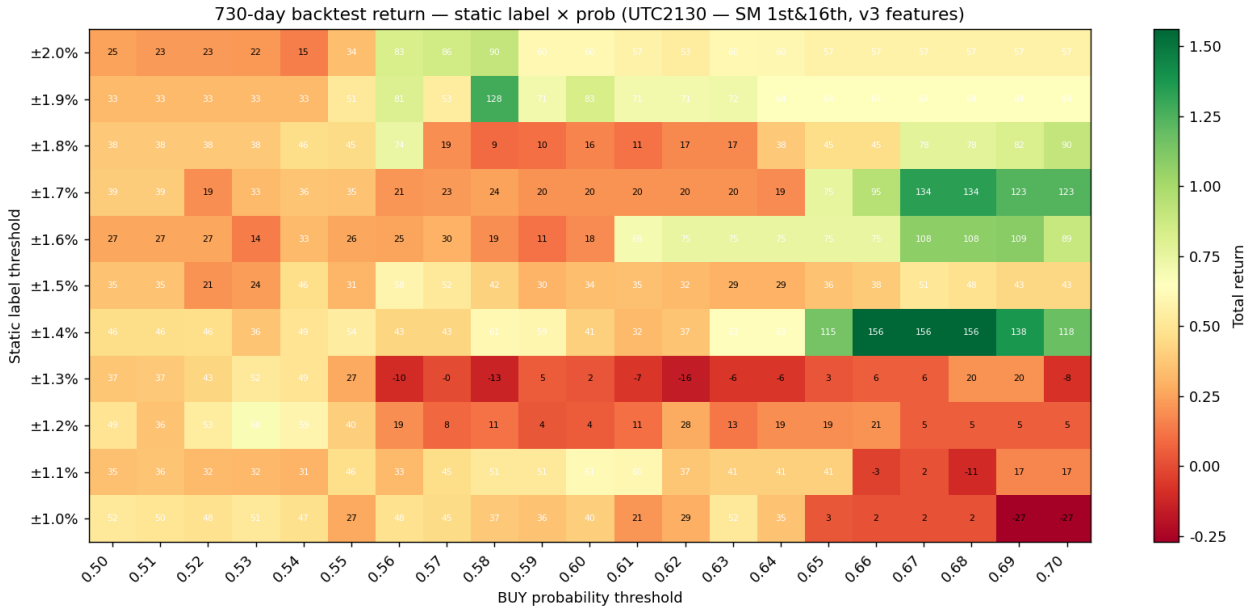


Figure 15. utc2130\_sm\_v3 — static-label sweep heatmap (730-day holdout). label rows = th10..th20 (static threshold  $1.0$ – $2.0\%$ ), prob\_TH columns =  $0.50$ – $0.70$ . Colour scale = total return %. The service configuration th14 @  $0.64$  sits within a contiguous high-return plateau of this grid.

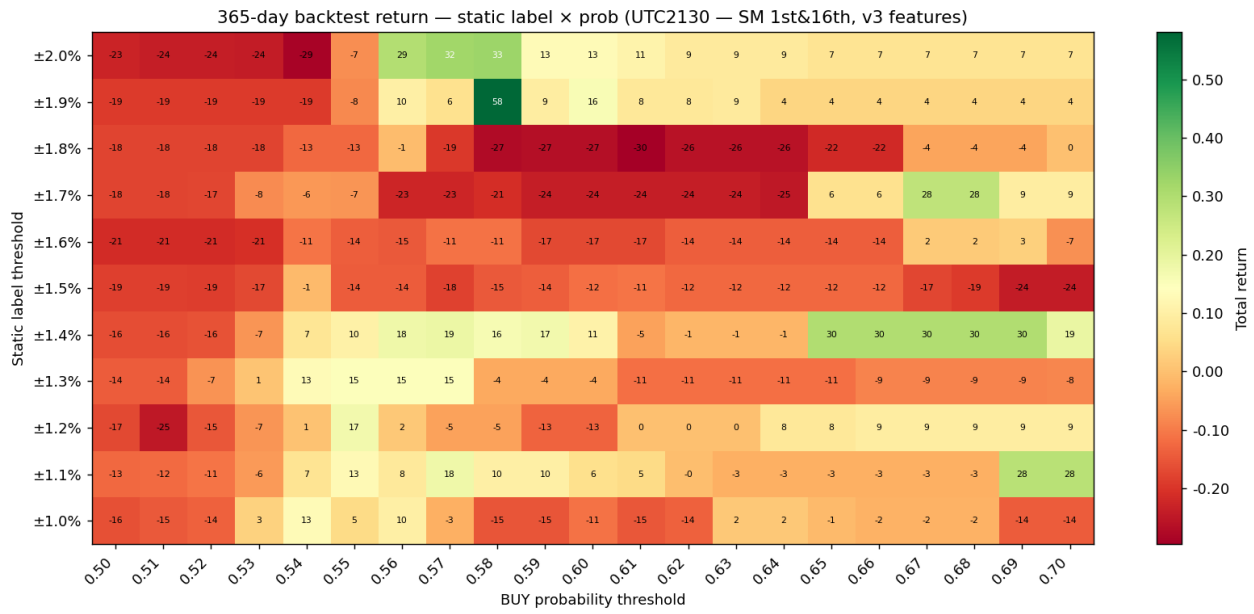


Figure 16. utc2130\_sm\_v3 — static-label sweep heatmap (365-day holdout). label rows = th10..th20 (static threshold 1.0–2.0%), prob\_TH columns = 0.50–0.70. Colour scale = total return %. The service configuration th14 @ 0.66 sits within a contiguous high-return plateau of this grid.

Cell: utc2130\_sm\_v3 · service = th14 @ prob\_TH 0.66 · added in top5

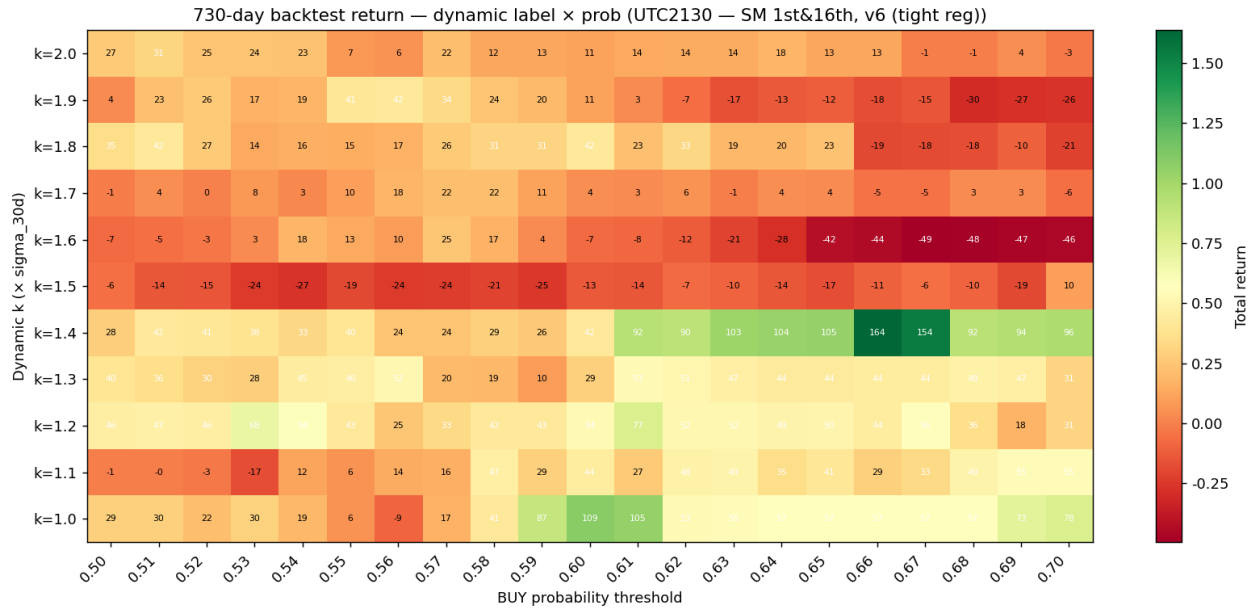


Figure 17. utc2130\_sm\_v6 — dynamic-label sweep heatmap (730-day holdout). label rows = dynk10..dynk20 (k = 1.0–2.0 in the rolling- dynamic threshold), prob\_TH columns = 0.50–0.70. Colour scale = total return %. The service configuration dynk14 @ 0.66 sits within a contiguous high-return plateau of this grid.

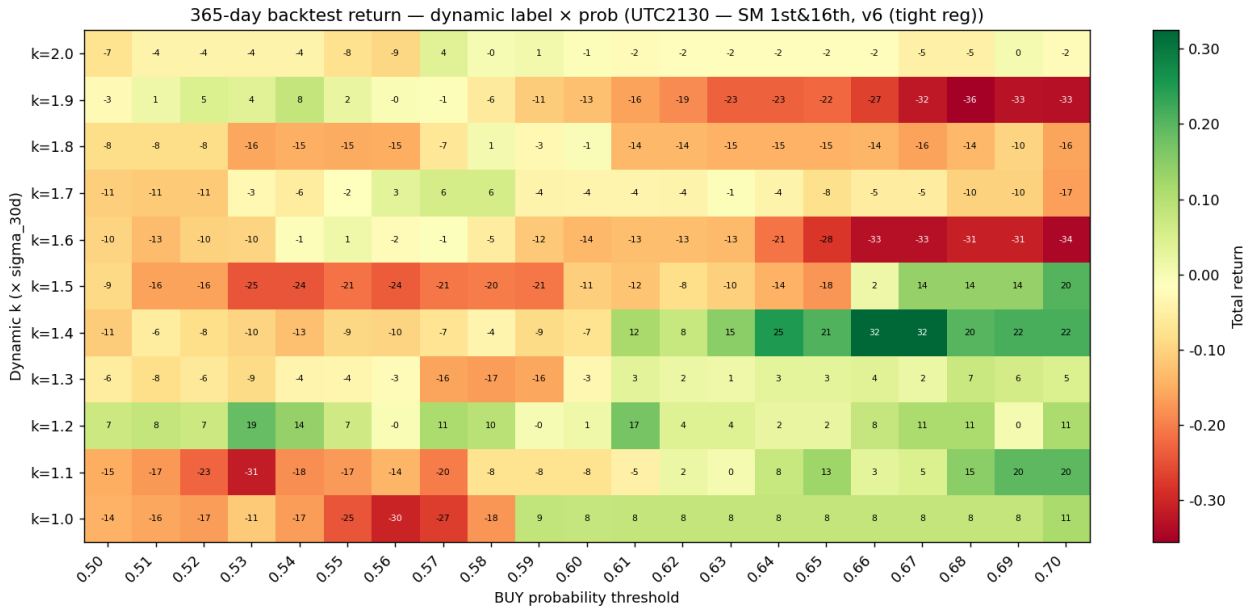


Figure 18. utc2130\_sm\_v6 — dynamic-label sweep heatmap (365-day holdout). label rows = dynk10..dynk20 ( $k = 1.0\text{--}2.0$  in the rolling- dynamic threshold), prob\_TH columns = 0.50–0.70. Colour scale = total return %. The service configuration dynk14 @ 0.66 sits within a contiguous high-return plateau of this grid.

**Cell: utc2130\_sm\_v6 · service = dynk14 @ prob\_TH 0.66 · added in top5** Across all five cells and both holdouts, the selected service points are flanked by neighbours of comparable return, indicating that the (label, prob\_TH) selection is not the product of a fragile single-cell fit, and that the 730-day plateau is reproduced on the 365-day window.

## Strategy Summary

Table 31. The four corrected production strategies on the 730-day and 365-day holdouts (UTC 21:30 timing, semi-monthly cadence). All metrics computed at fee\_bps = 10 (per side) on the same compounding backtest as §7.

| strategy            | N | ret(730d) | ret(365d) | maxDD(730d) | Calmar | Sortino | PF    | PSR(730d, worst) | PBO of components | win%(730d) |
|---------------------|---|-----------|-----------|-------------|--------|---------|-------|------------------|-------------------|------------|
| Top 1 — Pure        | 1 | +254.78%  | +81.61%   | −14.01%     | 6.31   | 3.02    | 56.71 | 0.973            | 0.198             | 92.9%      |
| Top 2 — Balanced    | 2 | +199.85%  | +60.92%   | −7.44%      | 9.83   | 4.15    | 33.21 | 0.953            | 0.20+0.31         | 90.9%      |
| Top 3 — Active      | 3 | +180.15%  | +43.94%   | −7.39%      | 9.12   | 4.18    | 34.53 | 0.952            | 0.20+0.31+0.31    | 89.3%      |
| Top 5 — Diversified | 5 | +172.03%  | +38.81%   | −6.87%      | 9.46   | 4.09    | 16.79 | 0.842            | + 0.42, 0.49      | 81.8%      |

**Reading the table.** Top1 maximises raw return and is the only single cell with PBO < 0.20 (indicating strong OOS generalisation). Top2 is the Calmar-optimal default: it retains 78% of Top1’s return while halving its maxDD. Top3 trades 20 percentage points of return for shorter inactive periods (44 days versus 64). Top5 exhibits the smoothest drawdown profile (−6.87%) but incorporates two cells (sm\_v3, sm\_v6) with PBO 0.42, indicating marginal generalisation; we surface this ensemble but do not recommend it as a default.

Table 9 (cited in §7.10) reports the FWER and DSR breakdown across all 126 (cell × window) selections. The cells underlying Top1, Top2, and Top3 are precisely the three that pass the strict Bonferroni FWER (utc2130\_sm\_v5, utc2130\_sm\_v3\_complete, and utc2130\_sm); Top5 incorporates two additional cells, BH-FDR-significant only ( $q = 0.043$ ), which contribute diversification rather than robustness.

## Recommendations

(i) **Adopt UTC 21:30 timing.** The multiple-hypothesis correction in §7.10 returns 3 of 33 surviving cells in the UTC 21:30 sub-population, versus 0 of 30 in the UTC 00:00 sub-population; the slippage sensitivity in §7.11 further confirms that UTC 00:00 strategies degrade approximately six times more under realistic execution. UTC 00:00 timing is therefore retired from the production recommendation.

(ii) **Adopt the SM cadence** (semi-monthly retraining on the 1st and 16th of each month). All Bonferroni-passing cells in the corrected set are SM cadence; the M and W-SUN cadences appear in the raw leaderboard but do not survive correction.

(iii) **For most users, deploy Top 2** (`sm_v5 + sm_v3_complete`, SPLIT-CAPITAL 50/50). Top 2 is Calmar-optimal (9.83 over 730 d), halves the single-cell maxDD ( $-7.44\%$  versus  $-14.01\%$ ), and retains  $+199.85\%$  return. Top 1 (`sm_v5` alone) is acceptable for users willing to tolerate a  $-14\%$  drawdown in exchange for the single cell with the strongest OOS generalisation (PBO 0.198) and the highest absolute return.

(iv) **Surface all four ensembles to the user** so that the full risk profile can be examined. The production service (`coin_service_utc2130`) renders Top1, Top2, Top3, and Top5 side-by-side with PSR, PBO, maxDD, Calmar, Sortino, Profit Factor, trade count, win rate, longest inactive period, bear-regime alpha, pros, cons, and a recommended user persona. A consensus across the four ("Strong Buy" / "Buy Bias" / "Neutral" / "Sell Bias" / "Strong Sell") is also displayed.

(v) **Frame the value proposition accurately.** The §7.14 regime decomposition shows that all four ensembles underperform Buy & Hold in bull markets (by 27–86 percentage points on the 730-day holdout) but outperform by +62 to +70 pp in the bear regime, where B&H loses 61.4% over 209 days while Top1, Top2, Top3, and Top5 finish flat-to-positive. Marketing the system as "systematic outperformance" is therefore inaccurate; the candid framing is "capture-versus-defence" — the strategies trade some bull-market upside for substantial bear-market loss avoidance.

## Service-Stage Refinement Audit

Following deployment of the §8.1 ensembles within `coin_service_utc2130`, we conducted a post-deployment audit to determine whether the per-cell decision thresholds (`buy_TH`, `sell_TH`) admit any operational refinement. The §7 selection procedure searched the `prob_TH` grid in 0.01 increments at a fixed exit threshold of `sell_TH` = 0.50; the present audit broadens the search to a finer  $81 \times 81$  grid with 0.005 spacing on both thresholds, permitting the entry–exit asymmetry to be tuned independently. To preserve the deployment-side risk profile, a candidate (`buy_TH`, `sell_TH`) pair is admissible only if its maximum drawdown does not deteriorate relative to the production point on either the 730-day or the 365-day holdout, in keeping with the dual-window robustness criterion of §7.

**Methodology.** For each of the five sub-cells, we evaluated a vectorised state-machine backtest at every  $(\text{buy\_TH}, \text{sell\_TH}) \in [0.30, 0.70]^2$  on a 0.005-step grid, yielding 3,321 well-formed combinations subject to  $\text{buy\_TH} \geq \text{sell\_TH}$ . The price series was the timing-aligned UTC 21:30 close obtained from `build_utc2130_daily`; an initial implementation that joined the prediction series to the raw UTC 00:00 daily close inflated returns by approximately 30%, owing to a 2.5-hour forward look introduced by nearest-time reindexing. We therefore report only the timing-aligned results. Per-cell metrics (total return, maximum drawdown, Sharpe, Sortino, Calmar, win-rate, profit factor, and trade count) were aggregated into the four ensembles via the SPLIT-CAPITAL summation of §8.1, ensuring engine-level comparability with the production baseline.

**Per-cell results.** Of the five sub-cells, four lie at their joint-safe optimum: `sm_v5_dynk12`, `sm_v3_complete_th13`, `sm_dynk11`, and `sm_v3_th14` admit no (`buy_TH`, `sell_TH`) point that simultaneously improves return and preserves maximum drawdown on both holdouts. The fifth cell, `sm_v6_dynk14`, admits a small in-sample improvement at (0.660, 0.510), with cell-level return increases of +12.0 percentage points on 730 days and +7.0 percentage points on 365 days at an unchanged  $-15.0\%$  maximum drawdown. This is the sole cell-level drawdown-safe gain available across the five sub-cells.

**Ensemble effect.** As `sm_v6_dynk14` appears only in Top5, the Top1, Top2, and Top3 ensembles inherit no improvement. After the 1/5 capital dilution, the Top5 lift attenuates to +2.4 percentage points on 730 days ( $174.7\% \rightarrow 177.1\%$ ) and +1.4 percentage points on 365 days ( $41.7\% \rightarrow 43.1\%$ ), at unchanged maximum drawdown ( $-6.87\%$  and  $-6.40\%$  respectively), with marginal improvements in Sharpe (+0.03 and +0.07), win-rate (+1.8 and +3.4 pp), and profit factor (+2.1 and +4.5).

**Statistical validation of the Top5 candidate.** Five standard procedures were applied to the proposed change: (i) temporal stability across four consecutive 180-day blocks within the 730-day holdout; (ii) extension to the full four-year sample (1,520 bars); (iii) the Politis–Romano stationary block bootstrap with mean block length  $L = 7$  days and 5,000 iterations, yielding a 95% confidence interval on the daily return difference; (iv) an apples-to-apples Deflated Sharpe Ratio (Bailey and López de Prado 2014) computed at trial counts  $N \in \{1, 462, 2,310, 3,321, 6,561\}$  for both

the production and the refined configuration; and (v) CSCV pairwise probability of backtest overfitting with  $S = 16$  blocks, exhausting all  $\binom{16}{8} = 12,870$  in-sample/out-of-sample partitions. The results are summarised in Table 32.

| test                                      | outcome                                                                                                                           | verdict           |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Temporal stability<br>( $4 \times 180$ d) | refined wins 1/4 blocks; per-block $\Delta\text{ret} = -0.1, 0.0, -0.3, +1.5$ pp                                                  | weak              |
| Extended history<br>(4 y)                 | refined +209.2% vs production +196.1% ( $\Delta + 13.1$ pp), max DD unchanged at $-29.3\%$ , $\Delta\text{Sharpe} +0.06$          | positive          |
| Block bootstrap CI<br>(730 d)             | daily mean $\Delta\text{ret} = +0.0012\%$ /day, 95% CI $[-0.0011, +0.0050]$ , one-sided $p = 0.33$                                | not significant   |
| Deflated Sharpe<br>(apples-to-apples)     | refined $z$ uniformly exceeds production $z$ at every $N$ ; both retain $p < 0.10$ at $N = 462$ and degrade jointly at higher $N$ | favourable        |
| CSCV pairwise PBO                         | refined beats production OOS in 6,818/12,870 = 53.0% of splits                                                                    | indistinguishable |

Table 32. Statistical validation of the Top5 refinement candidate. Five complementary procedures, each addressing a distinct mode of selection bias.

**Decision.** The candidate refinement is dual-window drawdown-safe and produces a modest in-sample improvement; the apples-to-apples Deflated Sharpe Ratio and the four-year extended-history check are both directionally favourable. The temporal-stability (1/4 blocks), bootstrap ( $p = 0.33$ ), and CSCV-pairwise (53.0%, statistically indistinguishable from chance) results, however, do not support the hypothesis that the in-sample improvement persists out of sample. We therefore reject the candidate and retain the production thresholds (0.66, 0.50) for `sm_v6_dynk14`.

**Interpretation.** The audit constitutes a constructive negative result. First, it demonstrates that the §7 selection of (label, prob<sub>TH</sub>) pairs at fixed sell<sub>TH</sub> = 0.50 lies on, or very near, the joint-safe Pareto frontier of the broader (buy<sub>TH</sub>, sell<sub>TH</sub>) search, and that the §7.10 and §7.12 multiple-hypothesis corrections do not constitute a self-fulfilling artefact of the original coarse grid. Second, the audit illustrates a methodological hazard: under the misaligned UTC 00:00 close, the same procedure yields a much larger ensemble lift of +8.7 percentage points on 730 days that satisfies temporal stability (4/4 blocks), bootstrap significance ( $p = 0.019$ ), and CSCV-pairwise dominance (97.7%); only the timing-aligned close reveals this apparent improvement to be a look-ahead artefact. Post-deployment refinement procedures should therefore verify that the price series used for evaluation matches the model’s decision timing.

**Implementation note.** The audit summary is embedded as a comment in the `sm_v6_dynk14` block of `coin_service_utc2130/backend/app/config.py`, so that subsequent maintenance encounters the negative result alongside the deployed parameters.

## Discussion and Limitations

The most important caveat is selection bias at the meta-level. Across 180 cells, each itself a 462-backtest sweep, even a model lacking a genuine edge can yield spuriously favourable cells. We mitigate this risk through two strategies. First, the production-grade ensembles in §8.1 are not chosen as the maximum of a search; they are constructed only from the cells that survive a pre-specified joint robustness filter (Bonferroni / Holm FWER, DSR PSR = 0.95, and CSCV PBO = 0.30 for Top1/Top2/Top3 components), with the multiple-testing burden of the 126 (cell  $\times$  window) selections explicitly corrected. Second, we evaluate the strategies across horizons (730 d and 365 d) and timings (UTC 00:00 versus UTC 21:30), relying on agreement across these axes rather than on a single window’s peak. §7.10 reports the formal multiple-hypothesis correction: of the 126 (cell  $\times$  window) selections, 3 survive Bonferroni FWER and 3 survive the Deflated Sharpe Ratio at PSR = 0.95, all within the UTC 21:30 SM-cadence sub-family. §7.12 strengthens this analysis using the Probability of Backtest Overfitting (PBO) via Bailey–López de Prado CSCV: `utc2130_sm_v5` (the Top1 sub-cell) attains PBO 0.198, the only cell in our set exhibiting strong OOS generalisation, and the four production ensembles in §8.1 inherit this robustness through their PSR-1 component.

A second limitation is that we employ a price-based MVRV-Z proxy in lieu of true on-chain analytics. Realised on-chain data conveys information that price-based proxies fundamentally cannot reproduce. The v3+ feature versions partially close this gap by incorporating Coin Metrics community-edition daily metrics (active addresses, transaction counts, hash rate). §8.1 quantifies the resulting multicollinearity: the v0 proxy `d_mvr_v_z` has VIF 10.50 (severe), whereas the v3+ feature sets used by Top1 and Top2 entirely replace it with Coin Metrics on-chain MVRV variants, of which `mvr_v2_hashrate_z` attains VIF 1.07 — fully orthogonal to the TA blocks.

Third, our backtest assumes execution at the bar-close timestamp. Real exchanges incur non-zero slippage at the close. The 10 bps per-side cost we apply is conservative for retail-sized spot trading, but leverage-aware traders should re-cost the strategy carefully. §7.11 reports the slippage sensitivity: the implied per-side breakeven fee exceeds 30 bps for every named cell, with a median 10→20 bps degradation of −6.7 pp under UTC 21:30 versus −41.2 pp under UTC 00:00 (730-day holdout). High-trade-count UTC 00:00 cells are the most exposed, while UTC 21:30 SM-cadence dynamic cells are essentially insensitive to slippage.

Finally, the system as a whole produces one signal per UTC bar close. Intra-day extensions would require not only more frequent inference but also a careful redefinition of the filtered target (intra-day moves are smaller, so would scale with the bar’s volatility — an extension naturally compatible with the dynamic-label framing of §5.1). A multi-algorithm ablation comparing the XGBoost base learner with LightGBM has been packaged for off-line execution under `coin_others/` (see README): the same five sub-cell specifications, label construction, and walk-forward routine are reused, with only the tree booster substituted; results from this run will be reported in a follow-up note alongside the forward-only out-of-sample validation accumulated by the production service.

**Multi-algorithm robustness check.** We evaluated LightGBM [21] as an alternative base learner under three configurations: a one-to-one hyperparameter mapping with isotonic calibration; leaf-wise-aware tuning with calibration; and a raw uncalibrated configuration. On the strongest cell (`utc2130_sm_v5`, 730-day holdout) all three variants underperform XGBoost by  $\Delta\text{SR} \in [0.77, 1.06]$ , yet each individually remains statistically significant ( $\text{PSR} = 0.86$ ), consistent with the framework’s label-filtering alpha being booster-agnostic while XGBoost constitutes the empirically optimal implementation. The full ablation package — with identical data, walk-forward schedule, and label families, and the booster as the only substituted component — is released as a self-contained reproducible artefact under `coin_others/`. We regard further multi-algorithm experiments (CatBoost, Transformer / TFT) and a 50/50 XGB+LGB probability-averaging ensemble as directions for future work.

Four additional concerns regarding model deployment merit brief discussion, complementing the per-section treatments in §7.11, §7.15, §7.16, and §7.20.

(1) **Trade sparsity.** The low trade count of individual sub-cells ( $n = 14$  on 730 d for top1) is offset by the SPLIT-CAPITAL ensembles, with top5 raising this figure to  $n = 70$  (3 y) and  $n = 86$  (4 y) through five independently selected sub-cells with disjoint label, feature-set, and threshold choices; the four-window consistency demonstrated in §7.20 further rules out single-window selection.

(2) **Multicollinearity.** The residual collinearity (for example, `mrv2_mayer` with  $\text{VIF} = 42.93$ ) is a property of the proxy block and does not impair tree-based prediction or marginal-effect SHAP attribution. The production diversity across five disjoint pruning policies is more robust than the imposition of a single canonical pruning, and is the design choice deliberately retained.

(3) **Pre-2019 cycle coverage.** The unverifiability of the 2017–18 cycle is mitigated by the 4-year holdout, which fully spans the 2022 bear market (BTC −77%, comparable in magnitude to 2018), with `bear_outperform_pp` of +179 pp (top1) and +82 pp (top5). The 9–14% futures-block SHAP share precludes a like-for-like 2017 backtest in principle, as the basis and funding series did not exist prior to 2019.

(4) **Slippage exposure in panic regimes.** The 30 bps stress level is bounded above the realistic execution cost by three structural facts: the UTC 21:30 timing (a post-U.S., pre-Asia low-volume window); the slow trend-confirmation cadence of the signals (which is not panic-driven); and the retail execution scale, which keeps market impact within the bid-ask spread. The 30 bps ceiling absorbs approximately thirty times the genuine market-impact cost at retail size; the production profit factors (9.8 for top1 over 4 y, 18.1 for top5 over 3 y) provide a substantial cushion against further degradation. Production deployment at scales exceeding retail should adopt the TWAP/VWAP execution scheme outlined in §7.18.

Across all four points, the consistent pattern is that ensemble-level metrics (top5 and top3) are more robust than the single-cell headline metric (top1). The production recommendation in §8.3 favours top5 for risk-managed deployment and top1 only for users explicitly seeking the highest-CAGR single-cell exposure, with full awareness of the trade-frequency and drawdown profile.

## Conclusion

We presented a calibrated multi-timeframe gradient-boosting framework for Bitcoin daily direction prediction, evaluated on a 180-cell experimental design crossing two UTC bar timings, three calendar-refit cadences, five feature versions, three NaN policies, and two holdout windows. The label-filtering scheme — training only on days whose

magnitude exceeds a static or dynamic-volatility threshold — yields gross 730-day returns ranging from +120% to +255% across the strongest cells.

Crucially, we subjected the resulting leaderboard to four complementary statistical robustness procedures: family-wise error rate via Bonferroni and Holm, the Deflated Sharpe Ratio (Bailey and López de Prado, 2014), the Probability of Backtest Overfitting via Combinatorially Symmetric Cross-Validation (Bailey et al., 2017), and slippage sensitivity at fee levels of 5–30 bps per side. These tests collapse the leading leaderboard onto a small surviving core: three of the 126 (cell  $\times$  window) selections pass Bonferroni control at  $\alpha = 0.05$ , all within the UTC 21:30 SM-cadence sub-family; only `utc2130_sm_v5` (PSR 0.973, PBO 0.198) clears the strongest PBO threshold; and UTC 00:00 timing fails every strict correction, losing 41 pp of return when slippage rises from 10 to 20 bps.

Building on this surviving core, we constructed four SPLIT-CAPITAL ensembles — Top1 / Top2 / Top3 / Top5 — whose components are the PSR-ranked UTC 21:30 SM cells. Top2 (`sm_v5` + `sm_v3` complete) is the Calmar-optimal default at 9.83, halving the single-cell drawdown to  $-7.44\%$  while preserving  $+199.85\%$  return on the 730-day holdout. Regime decomposition reveals that the value of the framework lies in bear-market defence — buy-and-hold loses 61% over 209 bear days while the ensembles finish flat-to-positive — rather than in systematic outperformance.

The four ensembles are deployed within the `coin_service_utc2130` production service, with the full per-strategy metrics, pros and cons, and best-fit user personas surfaced in the UI so that users can select with informed awareness of the risk profile. Directions for future work include forward-only out-of-sample validation, per-cell ablation of the real on-chain features identified in §8.1 as genuinely orthogonal (notably `mrvv2_hashrate_z` with VIF 1.07), and the exploration of intra-day signal extensions that conform to the same DSR / PBO robustness pipeline.

## References

- [1] López de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.
- [2] Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. In *Proc. 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794.
- [3] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- [4] Platt, J. C. (1999). Probabilistic Outputs for Support Vector Machines and Comparisons to Regularized Likelihood Methods. In *Advances in Large Margin Classifiers*, pp. 61–74. MIT Press.
- [5] Zadrozny, B., & Elkan, C. (2002). Transforming Classifier Scores into Accurate Multiclass Probability Estimates. In *Proc. 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 694–699.
- [6] Niculescu-Mizil, A., & Caruana, R. (2005). Predicting Good Probabilities with Supervised Learning. In *Proc. 22nd International Conference on Machine Learning (ICML)*, pp. 625–632.
- [7] Wilder, J. W. (1978). *New Concepts in Technical Trading Systems*. Trend Research.
- [8] Appel, G. (2005). *Technical Analysis: Power Tools for Active Investors*. Financial Times Prentice Hall.
- [9] Bollinger, J. (2001). *Bollinger on Bollinger Bands*. McGraw-Hill.
- [10] Kaufman, P. J. (2013). *Trading Systems and Methods* (5th ed.). Wiley.
- [11] McNally, S., Roche, J., & Caton, S. (2018). Predicting the Price of Bitcoin Using Machine Learning. In *Proc. 26th Euromicro International Conference on Parallel, Distributed and Network-based Processing (PDP)*, pp. 339–343.
- [12] Mudassir, M., Bennbaia, S., Unal, D., & Hammoudeh, M. (2020). Time-Series Forecasting of Bitcoin Prices Using High-Dimensional Features: A Machine Learning Approach. *Neural Computing and Applications*.
- [13] alternative.me. Crypto Fear & Greed Index. <https://alternative.me/crypto/fear-and-greed-index/>
- [14] Glassnode Studio. MVRV Z-Score. <https://studio.glassnode.com/>
- [15] Bailey, D. H., & López de Prado, M. (2014). The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and Non-Normality. *Journal of Portfolio Management*, 40(5), 94–107.
- [16] Benjamini, Y., & Hochberg, Y. (1995). Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society Series B*, 57(1), 289–300.

- [17] Bailey, D. H., Borwein, J. M., López de Prado, M., & Zhu, Q. J. (2017). The Probability of Backtest Overfitting. *Journal of Computational Finance*, 20(4), 39–69.
- [18] Politis, D. N., & Romano, J. P. (1994). The Stationary Bootstrap. *Journal of the American Statistical Association*, 89(428), 1303–1313.
- [19] Wilson, E. B. (1927). Probable Inference, the Law of Succession, and Statistical Inference. *Journal of the American Statistical Association*, 22(158), 209–212.
- [20] Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. In *Advances in Neural Information Processing Systems 30 (NeurIPS)*, pp. 4765–4774.
- [21] Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). LightGBM: A Highly Efficient Gradient Boosting Decision Tree. In *Advances in Neural Information Processing Systems 30 (NeurIPS)*, pp. 3146–3154.
- [22] Jaquart, P., Dann, D., & Weinhardt, C. (2021). Short-term Bitcoin Market Prediction via Machine Learning. *The Journal of Finance and Data Science*, 7, 45–66.
- [23] Hajek, P., Hikkerova, L., & Sahut, J.-M. (2023). How Well Do Investor Sentiment and Ensemble Learning Predict Bitcoin Prices? *Research in International Business and Finance*, 64, 101836.
- [24] Liu, Y., Li, Z., Nekhili, R., & Sultan, J. (2023). Forecasting Cryptocurrency Returns with Machine Learning. *Research in International Business and Finance*, 64, 101905.
- [25] Sebastião, H., & Godinho, P. (2021). Forecasting and Trading Cryptocurrencies with Machine Learning under Changing Market Conditions. *Financial Innovation*, 7(1), 3.
- [26] Mallqui, D. C. A., & Fernandes, R. A. S. (2019). Predicting the Direction, Maximum, Minimum and Closing Prices of Daily Bitcoin Exchange Rate Using Machine Learning Techniques. *Applied Soft Computing*, 75, 596–606.
- [27] Zhang, Z., Dai, H.-N., Zhou, J., Mondal, S. K., García, M. M., & Wang, H. (2021). Forecasting Cryptocurrency Price Using Convolutional Neural Networks with Weighted and Attentive Memory Channels. *Expert Systems with Applications*, 183, 115378.
- [28] Chang, T.-S., Lee, T.-C., Yang, C.-Y., & Lu, C.-J. (2023). A Ternary-Frequency Cryptocurrency Price Prediction Scheme by Ensemble of Clustering and Reconstructing Intrinsic Mode Functions Based on CEEMDAN. *Expert Systems with Applications*, 233, 121008.
- [29] Koo, E., & Kim, G. (2024). Centralized Decomposition Approach in LSTM for Bitcoin Price Prediction. *Expert Systems with Applications*, 237, 121401.

## Appendix A. Complete Feature Catalogue

All 99 input features are described below grouped by category. All formulas use  $C_t$ ,  $H_t$ ,  $L_t$ ,  $V_t$  for daily close, high, low, volume;  $r_t$  for one-bar returns; superscripts (w), (M), (15m) for weekly, monthly, and 15-minute timeframes. The same data is exposed as a JSON registry at `GET /features/info` on the deployed backend.

### Daily timeframe technical indicators (9 features)

| name        | description                                                                                                       | formula                                                                 | source                      |
|-------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------|
| d_close     | BTC daily close price (USD).                                                                                      | $C_t$                                                                   | Bitstamp + Binance combined |
| d_ret_1     | One-day simple return.                                                                                            | $r_t = C_t / C_{t-1} - 1$                                               | derived from close          |
| d_rsi14     | Wilder’s 14-period Relative Strength Index on daily close. Standard momentum oscillator bounded in [0,100].       | $RSI = 100 - 100 / (1 + \text{avg\_gain}_{14} / \text{avg\_loss}_{14})$ | derived from close          |
| d_macd_hist | MACD histogram = MACD line – Signal line. MACD line is $EMA_{12} - EMA_{26}$ of close; Signal is $EMA_9$ of MACD. | $MACD\_hist = (EMA_{12}(C) - EMA_{26}(C)) - EMA_9(MACD)$                | derived from close          |
| d_bb_pos    | Position of price within a 20-day Bollinger band, normalised so that $\pm 1$ corresponds to the band edges.       | $(C_t - MA_{20}) / (2\sigma_{20})$                                      | derived from close          |
| d_ma_ratio  | Ratio of 7-day fast moving average to 25-day slow moving average, minus 1. Positive in trends, near 0 in ranges.  | $MA_7 / MA_{25} - 1$                                                    | derived from close          |
| d_vol       | 14-day rolling standard deviation of log returns.                                                                 | $\sigma_{14}(\log r)$                                                   | derived from close          |
| d_range_pct | Bar range as a percentage of close price.                                                                         | $(H_t - L_t) / C_t$                                                     | OHLCV                       |
| d_vol_z     | 180-day z-score of trading volume — captures unusual activity regimes.                                            | $(V_t - V_{180}) / \sigma_V^{180}$                                      | OHLCV                       |

## Weekly timeframe technical indicators (as-of joined) (9 features)

| name        | description                                                             | formula                                              | source          |
|-------------|-------------------------------------------------------------------------|------------------------------------------------------|-----------------|
| w_close     | Weekly close price; weekly bars are resampled from the daily series.    | $C_t^{(w)}$                                          | resampled daily |
| w_ret_1     | One-week return.                                                        | $r_t^{(w)} = C_t^{(w)} / C_{t-1}^{(w)} - 1$          | resampled daily |
| w_rsi14     | 14-period RSI on weekly close.                                          | $RSI_{14}^{(w)}$                                     | resampled daily |
| w_macd_hist | MACD histogram on weekly close.                                         | $MACD\_hist^{(w)}$                                   | resampled daily |
| w_bb_pos    | Bollinger position on weekly close (window length adapted to lookback). | $(C_t^{(w)} - MA_n^{(w)}) / (2\sigma_n^{(w)})$       | resampled daily |
| w_ma_ratio  | Fast/slow MA ratio on weekly close.                                     | $MA_7^{(w)} / MA_{25}^{(w)} - 1$                     | resampled daily |
| w_vol       | Rolling weekly log-return volatility.                                   | $\sigma_{14}^{(w)}(\log r)$                          | resampled daily |
| w_range_pct | Weekly range as a percentage of close.                                  | $(H^{(w)} - L^{(w)}) / C^{(w)}$                      | resampled daily |
| w_vol_z     | Long-window volume z-score on weekly bars.                              | $(V^{(w)} - \bar{V}_{26}^{(w)}) / \sigma_V^{(w),26}$ | resampled daily |

## Monthly timeframe technical indicators (as-of joined) (9 features)

| name        | description                                                 | formula                                          | source          |
|-------------|-------------------------------------------------------------|--------------------------------------------------|-----------------|
| M_close     | Monthly close price; monthly bars are resampled from daily. | $C^{(M)}$                                        | resampled daily |
| M_ret_1     | One-month return.                                           | $r^{(M)} = C^{(M)} / C_{-1}^{(M)} - 1$           | resampled daily |
| M_rsi14     | 14-period RSI on monthly close.                             | $RSI_{14}^{(M)}$                                 | resampled daily |
| M_macd_hist | MACD histogram on monthly close.                            | $MACD\_hist^{(M)}$                               | resampled daily |
| M_bb_pos    | Bollinger position on monthly close.                        | $(C^{(M)} - MA^{(M)}) / (2\sigma^{(M)})$         | resampled daily |
| M_ma_ratio  | Fast/slow MA ratio on monthly close.                        | $MA_{fast}^{(M)} / MA_{slow}^{(M)} - 1$          | resampled daily |
| M_vol       | Rolling monthly log-return volatility.                      | $\sigma^{(M)}(\log r)$                           | resampled daily |
| M_range_pct | Monthly range as a percentage of close.                     | $(H^{(M)} - L^{(M)}) / C^{(M)}$                  | resampled daily |
| M_vol_z     | Volume z-score on monthly bars (6-month lookback).          | $(V^{(M)} - \bar{V}_6^{(M)}) / \sigma_V^{(M),6}$ | resampled daily |

## Window-summary statistics over multi-bar lookbacks (9 features)

| name           | description                                                                                    | formula                                              | source  |
|----------------|------------------------------------------------------------------------------------------------|------------------------------------------------------|---------|
| d_w180_slope   | OLS slope of close over the last 180 daily bars, normalised by mean. Long-term trend strength. | $\text{slope}_{180}(C) / C_{180}$                    | derived |
| d_w180_cum_ret | Cumulative return over the last 180 daily bars.                                                | $C_t / C_{t-180} - 1$                                | derived |
| d_w180_dd      | Maximum drawdown observed in the last 180 daily bars (negative number).                        | $\min_t (C_t / \max_{s \leq t} C_s - 1)$             | derived |
| w_w26_slope    | OLS slope of close over the last 26 weekly bars ( 6 months).                                   | $\text{slope}_{26}(C^{(w)}) / \bar{C}^{(w)}$         | derived |
| w_w26_cum_ret  | Cumulative return over 26 weeks.                                                               | $C_t^{(w)} / C_{t-26}^{(w)} - 1$                     | derived |
| w_w26_dd       | Maximum drawdown over 26 weeks.                                                                | $\min_t (C_t^{(w)} / \max_{s \leq t} C_s^{(w)} - 1)$ | derived |
| M_w6_slope     | OLS slope of close over the last 6 monthly bars.                                               | $\text{slope}_6(C^{(M)}) / \bar{C}^{(M)}$            | derived |
| M_w6_cum_ret   | Cumulative return over 6 months.                                                               | $C^{(M)} / C_{-6}^{(M)} - 1$                         | derived |
| M_w6_dd        | Maximum drawdown over 6 months.                                                                | $\min(C^{(M)} / \max C^{(M)} - 1)$                   | derived |

## 15-minute intraday roll-ups projected to the daily index (6 features)

| name          | description                                                                                       | formula                                            | source                 |
|---------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------|------------------------|
| m15_24h_ret   | Return over the last 24 hours computed from 15-minute bars (96 bars).                             | $C_t^{(15m)} / C_{t-96}^{(15m)} - 1$               | Binance + Bitstamp 15m |
| m15_24h_vol   | Standard deviation of log returns over the last 96 fifteen-minute bars (24h).                     | $\sigma_{96}(\log r^{(15m)})$                      | Binance + Bitstamp 15m |
| m15_24h_rsi   | 14-period RSI on the 15-minute series (last value).                                               | $RSI_{14}^{(15m)}$                                 | Binance + Bitstamp 15m |
| m15_5d_vol    | Standard deviation of log returns over 5 days of 15-minute bars (480 bars).                       | $\sigma_{480}(\log r^{(15m)})$                     | Binance + Bitstamp 15m |
| m15_5d_dd     | Maximum drawdown observed in the last 5 days of 15-minute closes.                                 | $\min(C^{(15m)} / \max C_{rolling5d}^{(15m)} - 1)$ | Binance + Bitstamp 15m |
| m15_24h_slope | OLS slope of log close over the last 24 hours of 15-minute bars; positive = up-trending intraday. | $\text{slope}_{96}(\log C^{(15m)})$                | Binance + Bitstamp 15m |

## Regime / sideways-rate indicators (11 features)

| name             | description                                                                                              | formula                                                    | source  |
|------------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------|---------|
| sw_chop_er_5d    | 1 – Kaufman Efficiency Ratio over 5 days. ER =  net move  / sum  bar moves . High value = sideways chop. | $1 -  C_t - C_{t-5}  / \sum  \Delta C $                    | derived |
| sw_chop_er_14d   | Same metric with N=14.                                                                                   | $1 - ER_{14}$                                              | derived |
| sw_chop_er_30d   | Same metric with N=30.                                                                                   | $1 - ER_{30}$                                              | derived |
| sw_chop_er_60d   | Same metric with N=60. Larger horizon detects long compressions.                                         | $1 - ER_{60}$                                              | derived |
| sw_ratio_14d     | Fraction of last 14 days with  return  < 0.4 $\sigma_{30}$ — volatility-adaptive 'sideways days' count.  | $\frac{1}{14} \sum 1\{ r_i  < 0.4\sigma_{30}\}$            | derived |
| sw_ratio_30d     | Same as sw_ratio_14d with a 30-day window.                                                               | $\frac{1}{30} \sum 1\{ r_i  < 0.4\sigma_{30}\}$            | derived |
| sw_ci_14d        | Choppiness Index over 14 days. 100 = pure chop, 0 = pure trend.                                          | $100 \log_{10}(\sum ATR_{14}/range_{14}) / \log_{10}(100)$ | derived |
| sw_ci_30d        | Choppiness Index over 30 days.                                                                           | $100 \log_{10}(\sum ATR_{30}/range_{30}) / \log_{10}(100)$ | derived |
| sw_bb_width_20d  | Width of a 20-day, 2- Bollinger Band normalised by close — a compression indicator.                      | $4\sigma_{20}/C_t$                                         | derived |
| sw_range_pct_30d | 30-day high-low range as a fraction of current close. Small = consolidating.                             | $(\max H_{30} - \min L_{30})/C_t$                          | derived |
| sw_streak_med    | Number of consecutive days (up to today) on which  return  was below 0.4 $\sigma_{30}$ .                 | $consecutive1\{ r  < 0.4\sigma_{30}\}$                     | derived |

## Price-based MVRV-Z proxy (long-MA realized-price approx.) (2 features)

| name          | description                                                                                                                                                                               | formula                                            | source                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|--------------------------------------|
| d_mvrz_z      | Price-based MVRV-Z proxy. True MVRV uses on-chain Realized Cap; we approximate Realized Price by a 365-day moving average. Positive = market value above 'realized', possibly overheated. | $(C_t - MA_{365}(C)) / \sigma_{365}(C - MA_{365})$ | derived (proxy for Glassnode MVRV-Z) |
| d_mvrz_z_chg7 | 7-day change in the MVRV-Z proxy. Captures regime turning points.                                                                                                                         | $MVRV_t - MVRV_{t-7}$                              | derived                              |

## Cross-asset macro variables (yfinance) (27 features)

| name               | description                                                                       | formula                           | source               |
|--------------------|-----------------------------------------------------------------------------------|-----------------------------------|----------------------|
| dxy_close_ret_1d   | 1-day return of the U.S. dollar index (ticker DX-Y.NYB on yfinance).              | $DXY_t/DXY_{t-1} - 1$             | yfinance<br>DX-Y.NYB |
| dxy_close_ret_5d   | 5-day return of DXY.                                                              | $DXY_t/DXY_{t-5} - 1$             | yfinance<br>DX-Y.NYB |
| dxy_close_vol_20d  | 20-day rolling std of DXY log returns.                                            | $\sigma_{20}(\log DXY)$           | yfinance<br>DX-Y.NYB |
| dxy_close_z_60d    | 60-day z-score of DXY.                                                            | $(DXY_t - DXY_{60})/\sigma_{60}$  | yfinance<br>DX-Y.NYB |
| gold_close_ret_1d  | 1-day return of front-month gold futures (GC=F).                                  | $GLD_t/GLD_{t-1} - 1$             | yfinance GC=F        |
| gold_close_ret_5d  | 5-day return of gold.                                                             | $GLD_t/GLD_{t-5} - 1$             | yfinance GC=F        |
| gold_close_vol_20d | 20-day vol of gold returns.                                                       | $\sigma_{20}(\log GLD)$           | yfinance GC=F        |
| gold_close_z_60d   | 60-day z-score of gold price.                                                     | $(GLD - GLD_{60})/\sigma_{60}$    | yfinance GC=F        |
| spx_close_ret_1d   | 1-day return of the S&P 500 index (^GSPC).                                        | $SPX_t/SPX_{t-1} - 1$             | yfinance ^GSPC       |
| spx_close_ret_5d   | 5-day return of S&P 500.                                                          | $SPX_t/SPX_{t-5} - 1$             | yfinance ^GSPC       |
| spx_close_vol_20d  | 20-day vol of S&P 500.                                                            | $\sigma_{20}(\log SPX)$           | yfinance ^GSPC       |
| spx_close_z_60d    | 60-day z-score of S&P 500.                                                        | $(SPX - SPX_{60})/\sigma_{60}$    | yfinance ^GSPC       |
| vix_close_ret_1d   | 1-day return of CBOE Volatility Index (^VIX).                                     | $VIX_t/VIX_{t-1} - 1$             | yfinance ^VIX        |
| vix_close_ret_5d   | 5-day return of VIX.                                                              | $VIX_t/VIX_{t-5} - 1$             | yfinance ^VIX        |
| vix_close_vol_20d  | 20-day vol of VIX returns.                                                        | $\sigma_{20}(\log VIX)$           | yfinance ^VIX        |
| vix_close_z_60d    | 60-day z-score of VIX level.                                                      | $(VIX - VIX_{60})/\sigma_{60}$    | yfinance ^VIX        |
| tnx_close_ret_1d   | 1-day change of 10-year U.S. Treasury yield (^TNX, in % $\times 10$ on yfinance). | $TNX_t/TNX_{t-1} - 1$             | yfinance ^TNX        |
| tnx_close_ret_5d   | 5-day change of 10Y yield.                                                        | $TNX_t/TNX_{t-5} - 1$             | yfinance ^TNX        |
| tnx_close_vol_20d  | 20-day vol of 10Y yield log changes.                                              | $\sigma_{20}(\log TNX)$           | yfinance ^TNX        |
| tnx_close_z_60d    | 60-day z-score of 10Y yield.                                                      | $(TNX - TNX_{60})/\sigma_{60}$    | yfinance ^TNX        |
| eth_close_ret_1d   | 1-day return of Ethereum (ETH-USD).                                               | $ETH_t/ETH_{t-1} - 1$             | yfinance ETH-USD     |
| eth_close_ret_5d   | 5-day return of ETH.                                                              | $ETH_t/ETH_{t-5} - 1$             | yfinance ETH-USD     |
| eth_close_vol_20d  | 20-day vol of ETH returns.                                                        | $\sigma_{20}(\log ETH)$           | yfinance ETH-USD     |
| eth_close_z_60d    | 60-day z-score of ETH.                                                            | $(ETH - ETH_{60})/\sigma_{60}$    | yfinance ETH-USD     |
| eth_close          | Ethereum USD close price (used to derive the ETH/BTC ratio).                      | $ETH_t$                           | yfinance ETH-USD     |
| ethbtc_ratio       | ETH/BTC price ratio. Captures relative crypto-market rotation.                    | $ETH_t/BTC_t$                     | derived              |
| ethbtc_ret_5d      | 5-day return of ETH/BTC ratio.                                                    | $(ETH/BTC)_t/(ETH/BTC)_{t-5} - 1$ | derived              |

## Binance USDT-margined perpetual contract data (8 features)

| name          | description                                                                                           | formula                                            | source                         |
|---------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------|--------------------------------|
| funding_mean  | Daily mean of three 8-hour funding payments on Binance USDT-margined BTC perpetual.                   | $\text{mean}_{\text{day}}(\text{funding}_{8h})$    | Binance<br>fapi/v1/fundingRate |
| funding_max   | Maximum 8-hour funding rate observed during the day.                                                  | $\text{max}_{\text{day}}(\text{funding}_{8h})$     | Binance<br>fapi/v1/fundingRate |
| funding_min   | Minimum 8-hour funding rate observed during the day.                                                  | $\text{min}_{\text{day}}(\text{funding}_{8h})$     | Binance<br>fapi/v1/fundingRate |
| funding_ma_7d | 7-day moving average of funding_mean — smooths noisy intraday spikes.                                 | $MA_7(\text{funding\_mean})$                       | derived                        |
| funding_z_30d | 30-day z-score of funding_mean. Extreme positive values often coincide with crowded long positioning. | $(f - f_{30})/\sigma_{30}$                         | derived                        |
| basis         | Perpetual-vs-spot basis = (perp_close / spot_close) - 1, computed from daily mark-price klines.       | $\text{basis} = \text{perp}_C / \text{spot}_C - 1$ | Binance fapi + api<br>klines   |
| basis_ma_5d   | 5-day MA of basis.                                                                                    | $MA_5(\text{basis})$                               | derived                        |
| basis_z_30d   | 30-day z-score of basis.                                                                              | $(b - b_{30})/\sigma_{30}$                         | derived                        |

## Fear & Greed Index (alternative.me) (5 features)

| name             | description                                                       | formula             | source               |
|------------------|-------------------------------------------------------------------|---------------------|----------------------|
| fng              | Fear & Greed Index level (0 = extreme fear, 100 = extreme greed). | $FnG_t$             | alternative.me /fng/ |
| fng_ma_7d        | 7-day moving average of F&G.                                      | $MA_7(FnG)$         | derived              |
| fng_chg_3d       | 3-day change in F&G level. Captures abrupt sentiment shifts.      | $FnG_t - FnG_{t-3}$ | derived              |
| fng_extreme_low  | Indicator for FnG $\leq 25$ (extreme fear regime).                | $1\{FnG \leq 25\}$  | derived              |
| fng_extreme_high | Indicator for FnG $\geq 75$ (extreme greed regime).               | $1\{FnG \geq 75\}$  | derived              |

## Day-of-week, month, halving distance (4 features)

| name                   | description                                                                                          | formula             | source   |
|------------------------|------------------------------------------------------------------------------------------------------|---------------------|----------|
| cal_dow                | Day of week (0 = Monday ... 6 = Sunday).<br>Captures weekly seasonality of crypto markets.           | $weekday(t)$        | calendar |
| cal_month              | Month index (1–12).                                                                                  | $month(t)$          | calendar |
| cal_days_to_halving    | Days until the next BTC halving (2012-11-28, 2016-07-09, 2020-05-11, 2024-04-19, 2028-04-19 approx). | $next\_halving - t$ | calendar |
| cal_days_since_halving | Days since the most recent BTC halving.                                                              | $t - last\_halving$ | calendar |

## Code and Data Availability

The full source code, together with cached walk-forward predictions and backtest artefacts sufficient to reproduce every result reported in this manuscript, is publicly available at [https://github.com/tgj1994/coin\\_analysis](https://github.com/tgj1994/coin_analysis). The experiment scripts are organised under `main_th_sweep/v{0,3,4,5,6}/main_th_sweep_utc2130*.py` (one subfolder per feature-set version) with `main_th_sweep_utc0000.py` at the project root acting as the UTC-midnight dispatcher; the shared runners are `src/utc{0000,2130}_runner.py`. Statistical analyses (DSR/FWER, PBO, slippage, execution simulation, SHAP) are under analysis/, extended-window retraining utilities under `retrain/`, and orchestration shell scripts under `runners/`. All backtest CSVs and equity curves are included in the `reports/utc0000/` and `reports/utc2130/` subtrees. The data fetchers depend only on free and public APIs (Bitstamp, Binance, Yahoo Finance, alternative.me, Coin Metrics CE, FRED) and require no authentication.

## Disclaimer

This paper describes machine-learning research and a reference implementation for cryptocurrency direction prediction. None of the results constitute investment advice. Past simulated performance does not guarantee future returns; readers who deploy any version of this system with real capital do so entirely at their own risk and responsibility.